INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY (ICDFA)

Case Study 4 - Investigating Rhino Possession Evidence

Lab Title:	**Individual recovered-media and network-traffic investigation**
Student Name:	**Yusuf Oluwaseun Dada**
Student ID:	**2025/FWSD/11339**
Course Code:	**SBT-DF204**
Course Name:	**Computer Forensics Case Studies**

Instructor: Mr. Aminu Idris, AMCPN

Date: 26/10/2026

1. Case Scenario

This case study is a fictional training investigation. I was supplied with an evidence package consisting of a seized USB disk image and three network traces showing FTP and HTTP activity, labelled rhino.log, rhino2.log, and rhino3.log. The investigative question is whether deleted files, hidden content, reconstructed network objects, and the timing of recorded transfers can be assembled into a single, defensible sequence of events.

This was conducted as an offline examination of the supplied training artefacts only. It did not involve, and does not authorise, any contact with any person, account, system, host, or service named within the material.

2. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux VM	Run the analysis.
Disk-image tools	The Sleuth Kit (mmls, fsstat, fls, icat)	Review media layout and file-system records.
File recovery	PhotoRec / foremost	Carve deleted or unallocated files.
Steganography	Course-supplied steganography tool (e.g. steghide, per the teaching deck)	Detect and, if justified, recover embedded content.
Archive tools	unzip, 7z, course-approved password-recovery tool	Validate and, only if authorised, open a protected archive.
Packet analysis	Wireshark	Reconstruct and validate FTP/HTTP objects.
Hashing	sha256sum	Preserve evidence integrity.

3. Case Folder Set up and Preservation of the the Originals

```
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/SBT-DF204-Case4/{evidence,working,exported,reports,screenshots}
cd ~/SBT-DF204-Case4
pwd
/home/kali/SBT-DF204-Case4
```

```
┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/USB_image/DFRWS2005-RODEO.zip

┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ unzip DFRWS2005-RODEO.zip
Archive:  DFRWS2005-RODEO.zip
  inflating: RHINOUSB.dd
  inflating: rhino.log
  inflating: rhino2.log
  inflating: rhino3.log
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# openssl dgst -md5 rhino.log
MD5(rhino.log)= c0d0093eb1664cd7b73f3a5225ae3f30

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# openssl dgst -md5 rhino2.log
MD5(rhino2.log)= cd21eaf4acfb50f71ffff857d7968341

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# openssl dgst -md5 rhino3.log
MD5(rhino3.log)= 7e29f9d67346df25faaf18efcd95fc30

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54
```

```
┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54

┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ dd if=RHINOUSB.dd of=RHINOUSB_my_copy_.dd
506848+0 records in
506848+0 records out
259506176 bytes (260 MB, 247 MiB) copied, 3.3561 s, 77.3 MB/s
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# sha256sum RHINOUSB.dd RHINOUSB_my_copy_.dd
ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65  RH
INOUSB.dd
ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65  RH
INOUSB_my_copy_.dd
```

4. Evidence Acquisition and Integrity

Field	Disk Image	FTP/HTTP Trace(s)
Original file name	RHINOUSB.dd	rhino.log, rhino2.log, rhino3.log
Source	DFRWS 2005 “Rodeo” challenge dataset, via course GitHub mirror	rhino.log, rhino2.log, rhino3.log
Byte size	259,506,176 bytes (260 MB / 247 MiB)	
SHA-256 (original)	ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65	rhino.log: c0d0093eb1664cd7b73f3a5225ae3f30 rhino2.log: cd21eaf4acfb50f71ffff857d7968341 rhino3.log: 7e29f9d67346df25faaf18efcd95fc30
SHA-256 (working copy)	ce550424200a997c61b413941c8ef4df9619a2f96579674952294a176a32be65	rhino.log: c0d0093eb1664cd7b73f3a5225ae3f30 rhino2.log: cd21eaf4acfb50f71ffff857d7968341 rhino3.log: 7e29f9d67346df25faaf18efcd95fc30
Working-copy method	dd if=RHINOUSB.dd of=RHINOUSB_my_copy_.dd — 506,848+0 records in, 506,848+0 records out, 259,506,176 bytes copied, 3.3561 s, 77.3 MB/s	same
Tools and versions	The Sleuth Kit: ver 4.14.0 PhotoRec 7.2, Data Recovery Utility, February 2024 (Christophe GRENIER)	Wireshark 4.6.6.
Analysis workstation	kali	

5. Part A - Examine the Disk Image Before Recovery

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# fdisk -l RHINOUSB.dd
Disk RHINOUSB.dd: 247.48 MiB, 259506176 bytes, 506848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x00000000
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# parted RHINOUSB.dd
GNU Parted 3.8
Using /home/kali/SBT-DF204-Case4/RHINOUSB.dd
Welcome to GNU Parted! Type 'help' to view a list of commands.

(parted) print
Model:  (file)
Disk /home/kali/SBT-DF204-Case4/RHINOUSB.dd: 260MB
Sector size (logical/physical): 512B/512B
Partition Table: loop
Disk Flags:

Number  Start  End    Size   File system  Flags
 1      0.00B  260MB  260MB  fat16


(parted) quit
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# fsstat -b 512 -o 0 RHINOUSB.dd
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT16

OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16

Sectors before file system: 0

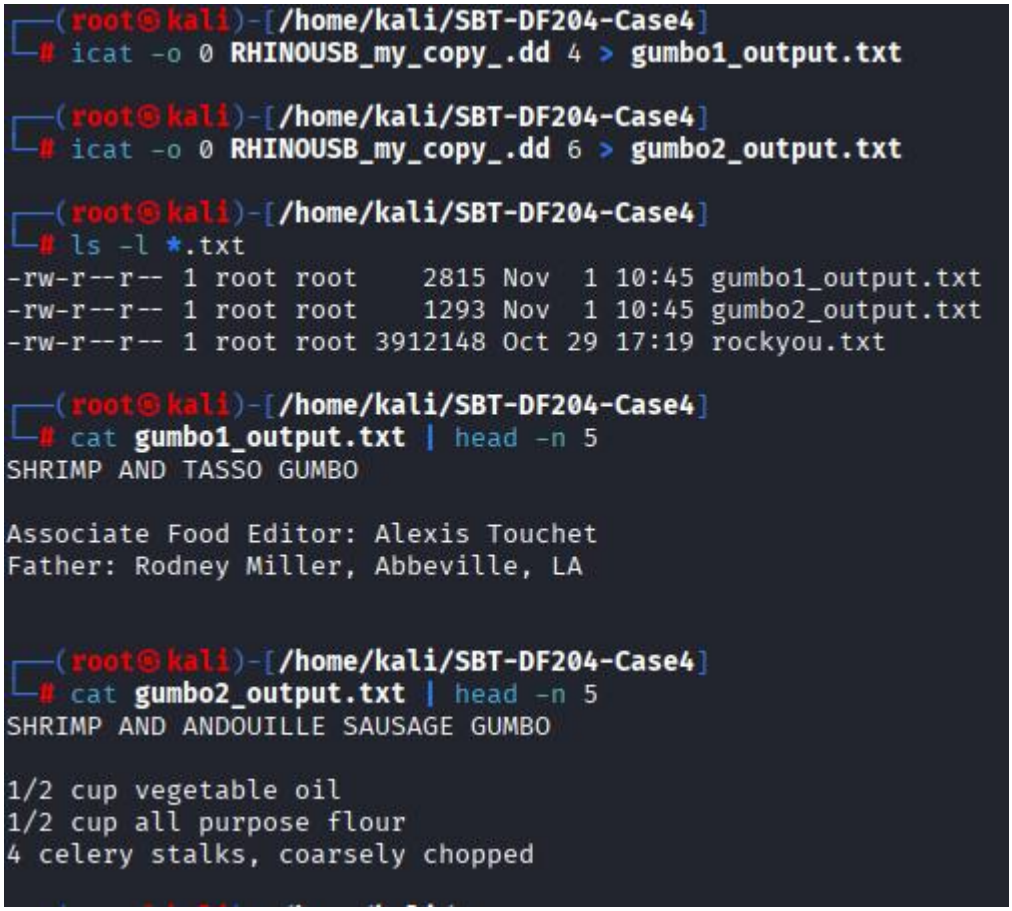
File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847

METADATA INFORMATION
--------------------------------------------
Range: 2 - 8101622
Root Directory: 2

CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 63290

FAT CONTENTS (in sectors)
--------------------------------------------
529-536 (8) -> EOF
537-544 (8) -> EOF
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# fls -o 0 RHINOUSB.dd
r/r 4:  gumbo1.txt
r/r 6:  gumbo2.txt
v/v 8101619:    $MBR
v/v 8101620:    $FAT1
v/v 8101621:    $FAT2
V/V 8101622:    $OrphanFiles
```



Item	Findings
Partition layout (from mmls)	No MBR-defined partition boundary found by fdisk; parted print shows a single loop partition table with one FAT16 partition spanning the full 260 MB (0.00B-260MB)
File-system type (from fsstat)	FAT16, OEM Name mkdosfs, Volume ID 0x4092d9d1, no volume label. Sector size 512B, cluster size 4096B. Root directory: sectors 497–528. Cluster area: sectors 529–506840 (cluster range 2–63290).
Allocated directory/file records of interest	From fls -o 0 RHINOUSB.dd: two allocated regular files, **inode 4:** gumbo1.txt, **inode 6:** gumbo2.txt (both marked r/r, i.e. allocated). Also listed: virtual metadata entries $MBR (8101619), $FAT1 (8101620), $FAT2 (8101621), $OrphanFiles (8101622).
Deleted entries of interest (marked * in fls)	None found via fls's standard deletion markers; the volume shows no asterisk-flagged deleted entries. This does not rule out recoverable content, PhotoRec's raw carving

Item	Findings
	does not depend on directory metadata and may still recover data not visible here
Limits of file-system metadata	Metadata records (names, sizes, timestamps) can be incomplete or already overwritten for deleted entries; a listed deleted entry does not guarantee the file's content is still recoverable.
File-system layout	Total range 0-506847 sectors; FAT 0 at sectors 1-248; FAT 1 at sectors 249-496; Root Directory at sectors 497-528; Cluster Area 529-506840

6. Part B - Recover Deleted and Hidden Content

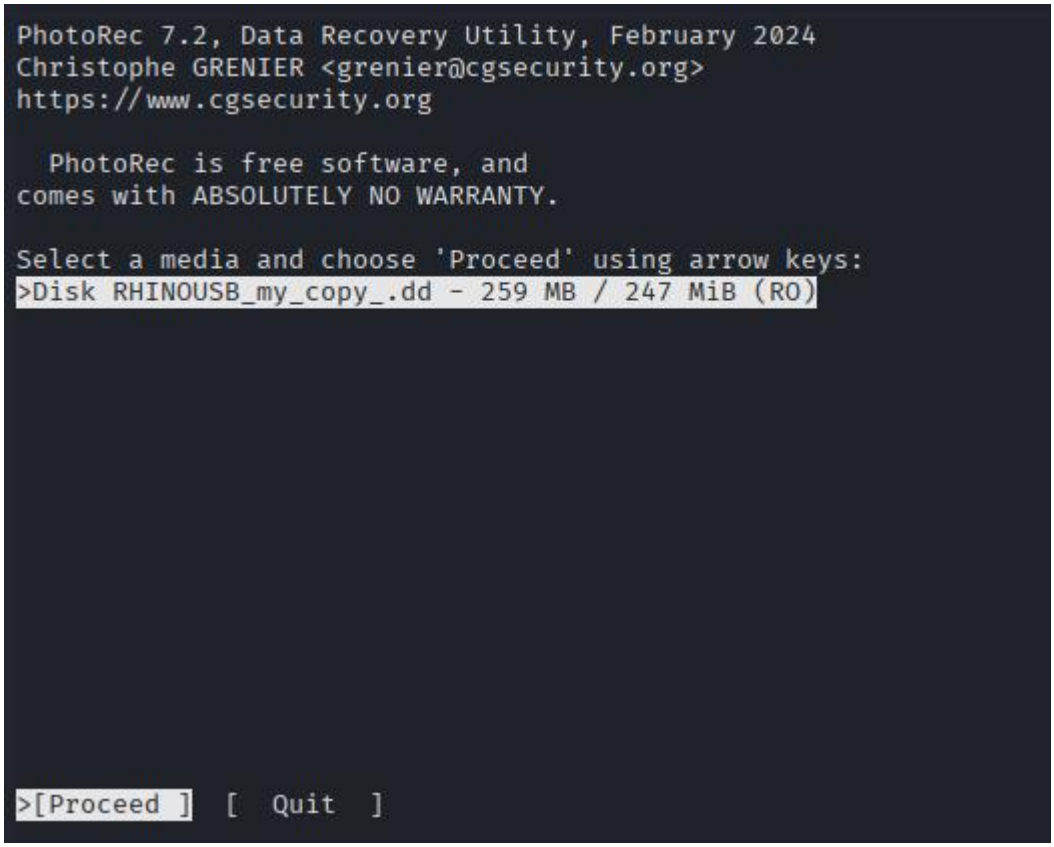
```
┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ mkdir -p ~/SBT-DF204-Case4/exported/photorec
photorec /log /d ~/SBT-DF204-Case4/exported/photorec "$IMG"
```

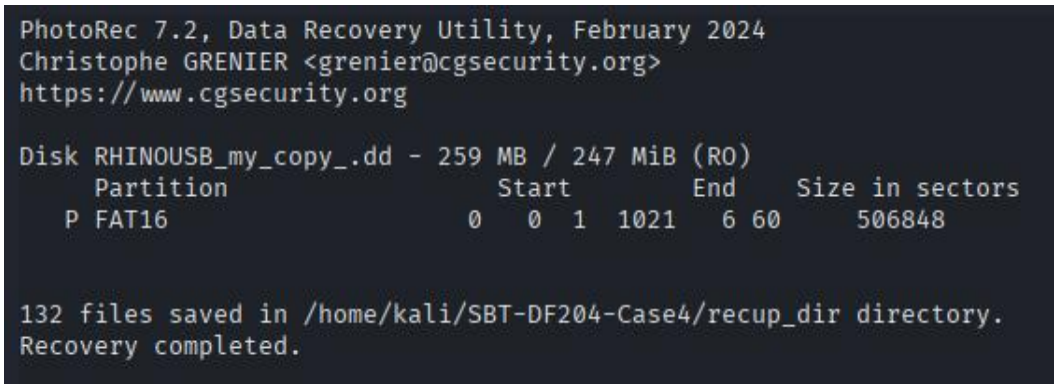
```
┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54

┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ dd if=RHINOUSB.dd of=RHINOUSB_my_copy_.dd
506848+0 records in
506848+0 records out
259506176 bytes (260 MB, 247 MiB) copied, 3.3561 s, 77.3 MB/s
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# openssl dgst -md5 RHINOUSB_my_copy_.dd
MD5(RHINOUSB_my_copy_.dd)= 80348c58eec4c328ef1f7709adc56a54
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# photorec RHINOUSB_my_copy_.dd
```


PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
PhotoRec is free software, and
comes with ABSOLUTELY NO WARRANTY.
Select a media and choose 'Proceed' using arrow keys:
>Disk RHINOUSB_my_copy_.dd - 259 MB / 247 MiB (RO)
>[Proceed] [Quit]

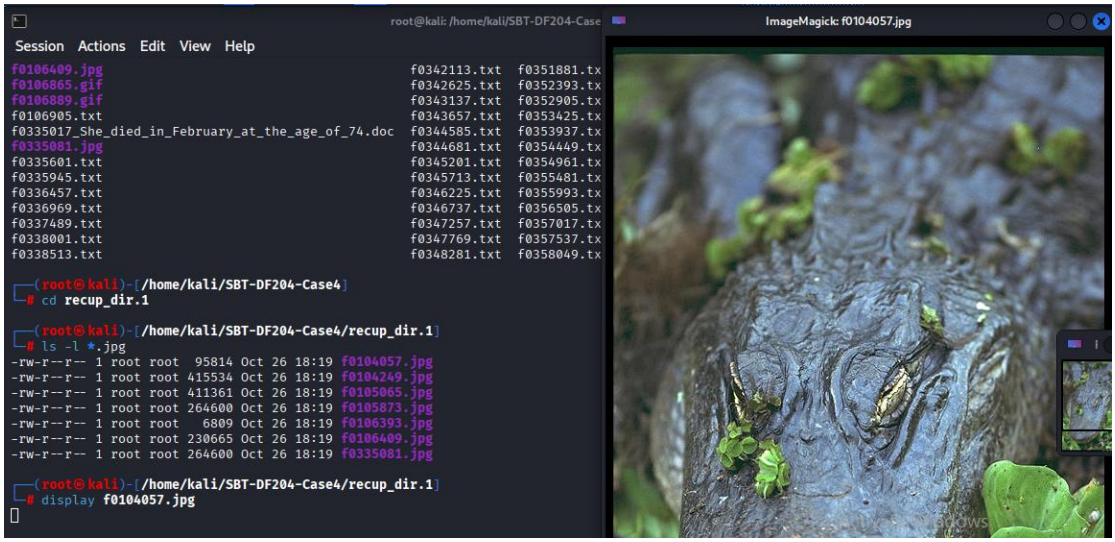

PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Disk RHINOUSB_my_copy_.dd - 259 MB / 247 MiB (RO)
Partition Start End Size in sectors
P FAT16 0 0 1 1021 6 60 506848
132 files saved in /home/kali/SBT-DF204-Case4/recup_dir directory.
Recovery completed.

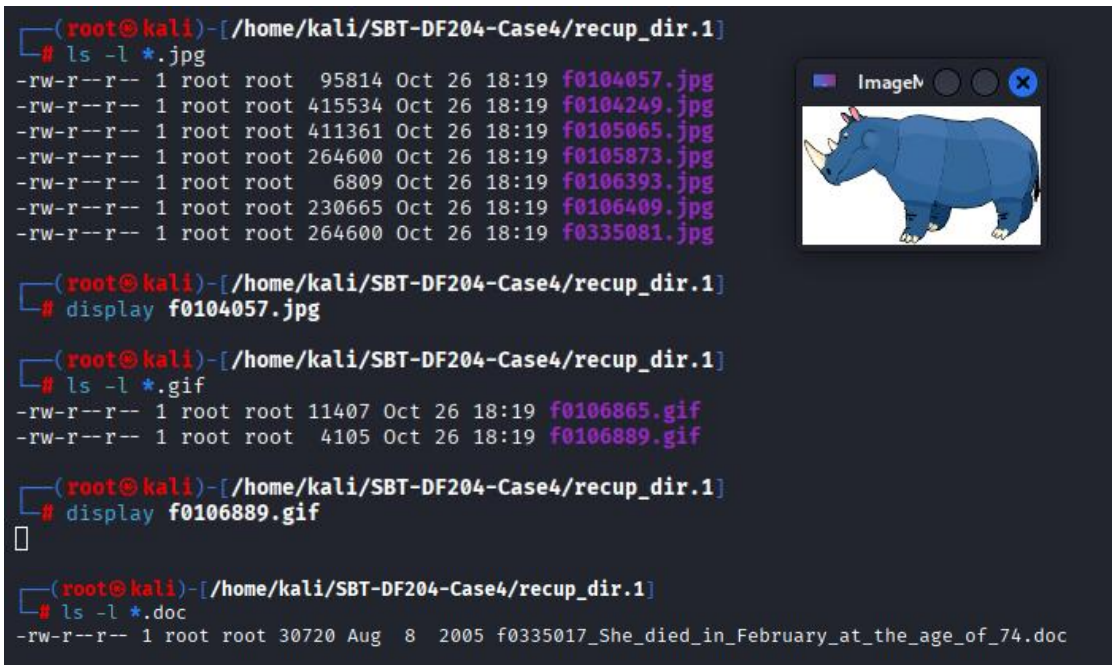
```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# ls -l
total 598052
-rw-r--r-- 1 root root    230566 Oct 28 18:52 contraband.zip
-rw-rw-r-- 1 kali kali   3614418 Oct 22 19:13 DFRWS2005-RODEO.zip
-rw-rw-r-- 1 kali kali   3614418 Oct 22 19:13 DFRWS2005-RODEO.zip.1
-rw-rw-r-- 1 kali kali    809088 Oct 30 14:18 DFRWS2005-RODEO.zip.2
-rw-rw-r-- 1 kali kali   3614418 Oct 30 14:18 DFRWS2005-RODEO.zip.3
drwxrwxr-x 2 kali kali      4096 Oct 23 17:52 evidence
drwxrwxr-x 4 kali kali      4096 Oct 23 18:26 exported
-rw-r--r-- 1 root root      2815 Nov  1 10:45 gumbo1_output.txt
-rw-r--r-- 1 root root      1293 Nov  1 10:45 gumbo2_output.txt
-rw-r--r-- 1 kali kali      7305 Oct 24 10:38 office
-rw-rw-r-- 1 kali kali     11462 Oct 23 18:30 photorec.log
-rw-r--r-- 1 root root     40960 Oct 26 18:19 photorec.se2
drwxr-xr-x 2 root root      4096 Oct 26 18:19 recup_dir.1
drwxrwxr-x 2 kali kali      4096 Oct 24 18:41 reports
-rw-rw-r-- 1 kali kali     65703 Oct 28 17:33 rhino1.jpg
-rw-r--r-- 1 root root    230665 Apr 26  2004 rhino2.jpg
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# ls recup_dir.1/
f0000545.txt                                f0339025.txt  f0348793.txt  f0358561.txt  f0368329.txt  f0378097.txt  f0387865.txt
f0104057.jpg                                f0339545.txt  f0349313.txt  f0359073.txt  f0368841.txt  f0378609.txt  f0388377.txt
f0104249.jpg                                f0340057.txt  f0349825.txt  f0359593.txt  f0369361.txt  f0379129.txt  f0388889.txt
f0105065.jpg                                f0340569.txt  f0350337.txt  f0360105.txt  f0369873.txt  f0379641.txt  f0389409.txt
f0105873.jpg                                f0341081.txt  f0350849.txt  f0360969.txt  f0370385.txt  f0380153.txt  f0389921.txt
f0106393.jpg                                f0341601.txt  f0351369.txt  f0361137.txt  f0370897.txt  f0380665.txt  f0390433.txt
f0106409.jpg                                f0342113.txt  f0351881.txt  f0361649.txt  f0371417.txt  f0381185.txt  f0390945.txt
f0106865.gif                                f0342625.txt  f0352393.txt  f0362161.txt  f0371929.txt  f0381697.txt  f0391465.txt
f0106889.gif                                f0343137.txt  f0352905.txt  f0362673.txt  f0372441.txt  f0382209.txt  f0391977.txt
f0106905.txt                                f0343657.txt  f0353425.txt  f0363193.txt  f0372953.txt  f0382721.txt  f0392489.txt
f0335017_She_died_in_February_at_the_age_of_74.doc  f0344585.txt  f0353937.txt  f0363705.txt  f0373473.txt  f0383241.txt  f0393001.txt
f0335081.jpg                                f0344681.txt  f0354449.txt  f0364217.txt  f0373985.txt  f0383753.txt  f0393737.txt
f0335601.txt                                f0345201.txt  f0354961.txt  f0364729.txt  f0374497.txt  f0384265.txt  f0394033.txt
f0335945.txt                                f0345713.txt  f0355481.txt  f0365249.txt  f0375009.txt  f0384777.txt  f0394545.txt
f0336457.txt                                f0346225.txt  f0355993.txt  f0365761.txt  f0375529.txt  f0385297.txt  f0395065.txt
f0336969.txt                                f0346737.txt  f0356505.txt  f0366273.txt  f0376041.txt  f0385809.txt  f0395577.txt
f0337489.txt                                f0347257.txt  f0357017.txt  f0366785.txt  f0376553.txt  f0386321.txt  f0396089.txt
f0338001.txt                                f0347769.txt  f0357537.txt  f0367305.txt  f0377353.txt  f0386833.txt  f0396601.txt
f0338513.txt                                f0348281.txt  f0358049.txt  f0367817.txt  f0377585.txt  f0387353.txt  report.xml
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cd recup_dir.1

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# ls -l *.jpg
-rw-r--r-- 1 root root  95814 Oct 26 18:19 f0104057.jpg
-rw-r--r-- 1 root root 415534 Oct 26 18:19 f0104249.jpg
-rw-r--r-- 1 root root 411361 Oct 26 18:19 f0105065.jpg
-rw-r--r-- 1 root root 264600 Oct 26 18:19 f0105873.jpg
-rw-r--r-- 1 root root   6809 Oct 26 18:19 f0106393.jpg
-rw-r--r-- 1 root root 230665 Oct 26 18:19 f0106409.jpg
-rw-r--r-- 1 root root 264600 Oct 26 18:19 f0335081.jpg
```


root@kali: /home/kali/SBT-DF204-Case
Session Actions Edit View Help
f0106409.jpg
f0106865.gif
f0106889.gif
f0106905.txt
f0335017_She_died_in_February_at_the_age_of_74.doc
f0335081.jpg
f0335601.txt
f0335945.txt
f0336457.txt
f0336969.txt
f0337489.txt
f0338001.txt
f0338513.txt
f0342113.txt f0351881.tx
f0342625.txt f0352393.tx
f0343137.txt f0352905.tx
f0343657.txt f0353425.tx
f0344585.txt f0353937.tx
f0344681.txt f0354449.tx
f0345201.txt f0354961.tx
f0345713.txt f0355481.tx
f0346225.txt f0355993.tx
f0346737.txt f0356505.tx
f0347257.txt f0357017.tx
f0347769.txt f0357537.tx
f0348281.txt f0358049.tx
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cd recup_dir.1
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# ls -l *.jpg
-rw-r--r-- 1 root root 95814 Oct 26 18:19 f0104057.jpg
-rw-r--r-- 1 root root 415534 Oct 26 18:19 f0104249.jpg
-rw-r--r-- 1 root root 411361 Oct 26 18:19 f0105065.jpg
-rw-r--r-- 1 root root 264600 Oct 26 18:19 f0105873.jpg
-rw-r--r-- 1 root root 6809 Oct 26 18:19 f0106393.jpg
-rw-r--r-- 1 root root 230665 Oct 26 18:19 f0106409.jpg
-rw-r--r-- 1 root root 264600 Oct 26 18:19 f0335081.jpg
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# display f0104057.jpg
ImageMagick: f0104057.jpg


┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# ls -l *.jpg
-rw-r--r-- 1 root root 95814 Oct 26 18:19 f0104057.jpg
-rw-r--r-- 1 root root 415534 Oct 26 18:19 f0104249.jpg
-rw-r--r-- 1 root root 411361 Oct 26 18:19 f0105065.jpg
-rw-r--r-- 1 root root 264600 Oct 26 18:19 f0105873.jpg
-rw-r--r-- 1 root root 6809 Oct 26 18:19 f0106393.jpg
-rw-r--r-- 1 root root 230665 Oct 26 18:19 f0106409.jpg
-rw-r--r-- 1 root root 264600 Oct 26 18:19 f0335081.jpg
ImageM
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# display f0104057.jpg
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# ls -l *.gif
-rw-r--r-- 1 root root 11407 Oct 26 18:19 f0106865.gif
-rw-r--r-- 1 root root 4105 Oct 26 18:19 f0106889.gif
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# display f0106889.gif
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/recup_dir.1]
└─# ls -l *.doc
-rw-r--r-- 1 root root 30720 Aug 8 2005 f0335017_She_died_in_February_at_the_age_of_74.doc

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cp recup_dir.1/f0104057.jpg steg/

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cp recup_dir.1/f0104249.jpg steg/

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cp recup_dir.1/f0105065.jpg steg/

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cp recup_dir.1/f0105873.jpg steg/

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cp recup_dir.1/f0335081.jpg steg/

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cd steg/

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg]
└─# ls
f0104057.jpg  f0104249.jpg  f0105065.jpg  f0105873.jpg  f0335081.jpg
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg]
└─# exiftool f0104057.jpg
ExifTool Version Number         : 13.55
File Name                       : f0104057.jpg
Directory                       : .
File Size                       : 96 kB
File Modification Date/Time     : 2026:10:27 14:11:43-04:00
File Access Date/Time           : 2026:10:27 14:11:42-04:00
File Inode Change Date/Time     : 2026:10:27 14:11:43-04:00
File Permissions                : -rw-r--r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : None
X Resolution                    : 0
Y Resolution                    : 0
Comment                         : copyright 2000 philg@mit.edu
Image Width                     : 528
Image Height                    : 792
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 528×792
Megapixels                      : 0.418
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# cd steg

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg]
└─# git clone https://github.com/poizan42/stegdetect
Cloning into 'stegdetect'...
remote: Enumerating objects: 419, done.
remote: Total 419 (delta 0), reused 0 (delta 0), pack-reused 419 (fr
om 1)
Receiving objects: 100% (419/419), 1.38 MiB | 124.00 KiB/s, done.
Resolving deltas: 100% (59/59), done.

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg]
└─# cd stegdetect

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# apt-get install dh-autoreconf
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Solving dependencies... Done
The following packages were automatically installed and are no longer required:
  chromium-sandbox gnome-accessibility-themes gnome-themes-extra libavfilter11 libavformat62 libboost-locale1.90.0 libboost-thread1.90.0
  libconfig++11 libconfig11 libflashrom1 libfmt10 libgeos3.14.1 libhdf4-0 libimath-3-1-29t64 libjim0.83 libkmlbase1t64 libkmldom1t64
  libkmlengine1t64 libmupdf27.0 libntfs-3g90 libperl5.40 libpoppler156 libpython3.13 libpython3.13-minimal libpython3.13-stdlib
  libswscale9 openssh-client-gssapi perl-modules-5.40 python3-mitmproxy-wireguard python3.13-tk rpcsvc-proto
Use 'sudo apt autoremove' to remove them.
The following additional packages will be installed:
```

Activate Windows

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# autoreconf -f -i
autoreconf: warning: autoconf input should be named 'configure.ac', not 'configure.in'
aclocal: warning: autoconf input should be named 'configure.ac', not 'configure.in'
aclocal: warning: autoconf input should be named 'configure.ac', not 'configure.in'
acinclude.m4:2: warning: underquoted definition of AC_C_LONG_LONG
acinclude.m4:2:   run info Automake 'Extending aclocal'
acinclude.m4:2:   or see https://www.gnu.org/software/automake/manual/automake.html#Extending-aclocal
configure.in:3: warning: AM_INIT_AUTOMAKE: two- and three-arguments forms are deprecated.
./lib/autoconf/general.m4:2490: AC_DIAGNOSE is expanded from...
aclocal.m4:429: AM_INIT_AUTOMAKE is expanded from...
configure.in:3: the top level
configure.in:4: warning: 'AM_CONFIG_HEADER': this macro is obsolete.
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# linux32 make
make  all-recursive
make[1]: Entering directory '/home/kali/SBT-DF204-Case4/steg/stegdetect'
Making all in jpeg-6b
make[2]: Entering directory '/home/kali/SBT-DF204-Case4/steg/stegdetect/jpeg-6b'
make[2]: *** No rule to make target 'all'.  Stop.
make[2]: Leaving directory '/home/kali/SBT-DF204-Case4/steg/stegdetect/jpeg-6b'
make[1]: *** [Makefile:638: all-recursive] Error 1
make[1]: Leaving directory '/home/kali/SBT-DF204-Case4/steg/stegdetect'
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# ./stegdetect -V
Stegdetect Version 0.6.1
```

```
[file.jpg ...]

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# mkdir /usr/local/share/stegbreak

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# cp rules.ini /usr/local/share/stegbreak/

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# ./stegbreak
Usage: ./stegbreak [-V] [-r <rules>] [-f <wordlist>] [-t <schemes>]

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# wget -q https://github.com/brannondorsey/naive-hashcat/releases/
^C

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# wget https://github.com/brannondorsey/naive-hashcat/releases/dow
--2026-11-01 15:55:18--  https://github.com/brannondorsey/naive-hash
Resolving github.com (github.com)... 140.82.121.4
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# ./stegbreak -f rockyou.txt ../*.jpg
Loaded 5 files...
../f0105065.jpg : jphide[v5](gator)
../f0104249.jpg : jphide[v5](gumbo)
zsh: segmentation fault  ./stegbreak -f rockyou.txt ../*.jpg
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# mkdir stegseekexe

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4/steg/stegdetect]
└─# cd stegseekexe

┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
]
└─# wget -q ftp://ftp.gwdg.de/pub/linux/misc/ppdd/jphs_05.zip

┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
]
└─# ls
jphs_05.zip

┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
]
└─# unzip jphs_05.zip
Archive:  jphs_05.zip
 extracting: jphs05.zip
 extracting: jphs05.zip.sig
```

```
┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
]
└─# wine jpseek.exe /home/kali/SBT-DF204-Case4/steg/f0105065.jpg r06
5.jpg

Welcome to jpseek Rev 0.51
 (c) 1998 Allan Latham <alatham@flexsys-group.com>
This program is freeware.
No charge is made for its use.
Use at your own risk. No liability accepted whatever happens.
Contains cryptogaphy which may be subject to local laws.

Passphrase:

┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
]
└─# display r065.jpg
```




```
┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
└─# wine jpseek.exe /home/kali/SBT-DF204-Case4/steg/f0104249.jpg r24
9.jpg

Welcome to jpseek Rev 0.51
 (c) 1998 Allan Latham <alatham@flexsys-group.com>
This program is freeware.
No charge is made for its use.
Use at your own risk. No liability accepted whatever happens.
Contains cryptogaphy which may be subject to local laws.

Passphrase:

┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
└─# display r249.jpg
```

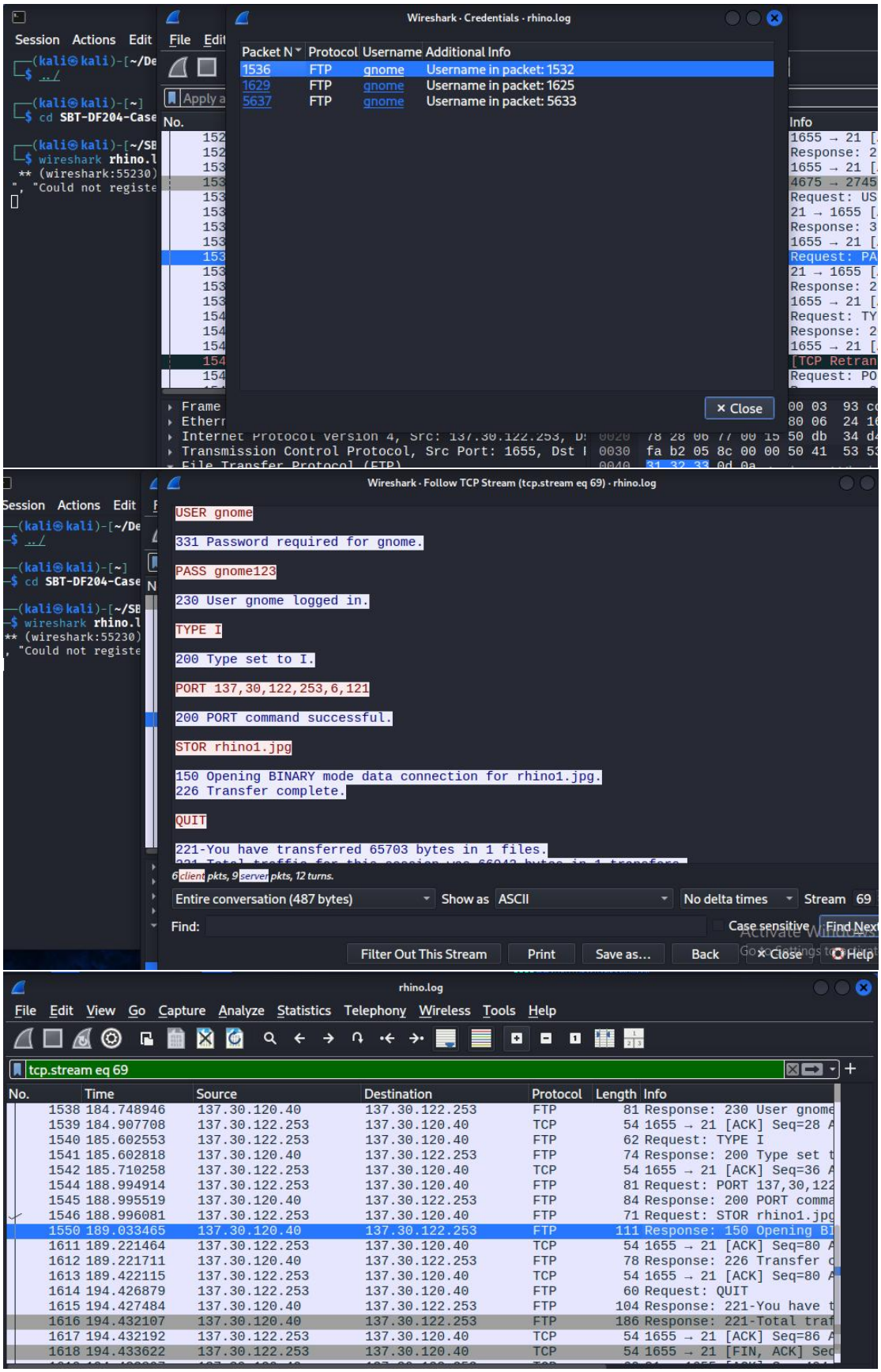


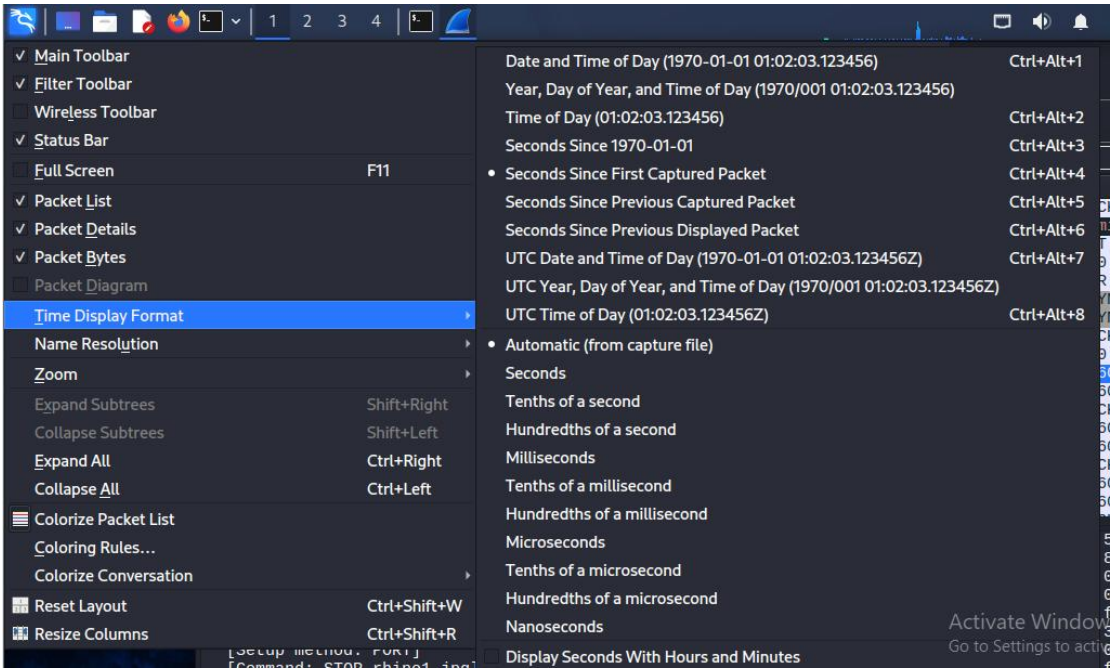

```
┌──(root㉿kali)-[/home/…/SBT-DF204-Case4/steg/stegdetect/stegseekexe
└─# openssl dgst -md5 r065.jpg r249.jpg
openssl dgst -sha256 r065.jpg r249.jpg
MD5(r065.jpg)= 63a39823f80b321c2dcd112158b55011
MD5(r249.jpg)= 87018ef0cfdb91e818d92efeb9c19338
SHA2-256(r065.jpg)= bc9c605372fdf9cc00faf17ae9ce0fdb6d26bb9db28cdff7
42d9ddbe1470ec16
SHA2-256(r249.jpg)= 6dbf775b8999a006ffa74c65127e5add64234184bd2f425b
2ad644e8c0d19345
```

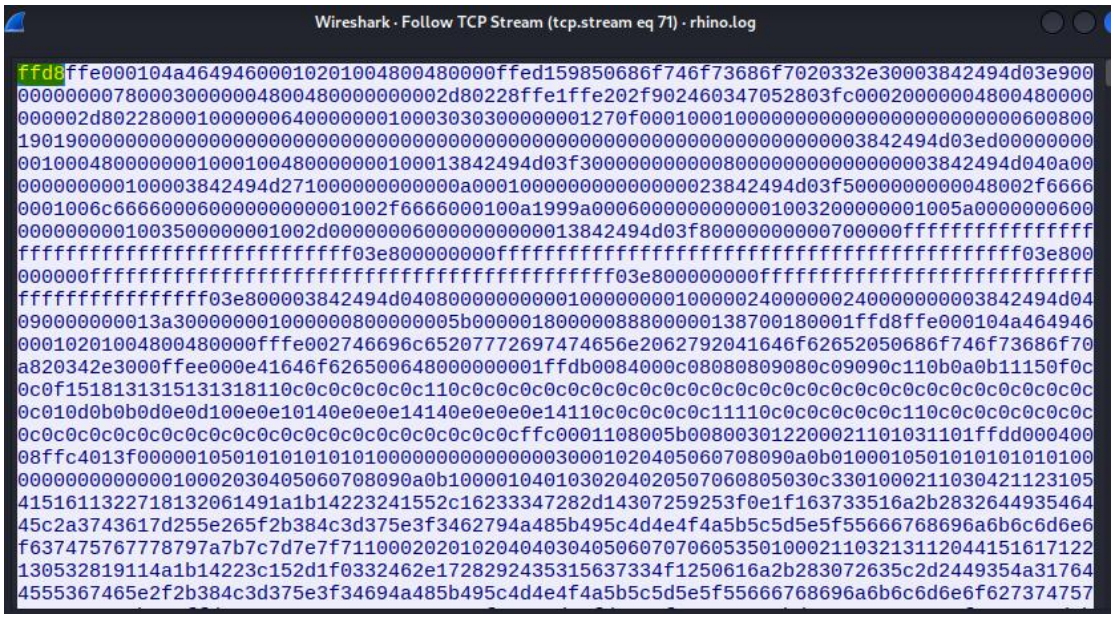
Item	Findings
Recovery method and scope selected	Two methods used: (1) icat to extract known allocated inodes directly by inode number; (2) PhotoRec, full image scan, output to recup_dir.1/
Output folder	recup_dir.1/ (PhotoRec); direct extracts via icat saved to working directory as gumbo1_output.txt, gumbo2_output.txt
Recovered file(s) of interest	132 files
Duplicate content confirmed	f0105873.jpg and f0335081.jpg, identical SHA-256, same image carved from two locations
Notable image metadata	f0104057.jpg contains embedded comment “copyright 2000 philg@mit.edu”; f0106393.jpg contains comment “LEAD Technologies Inc.

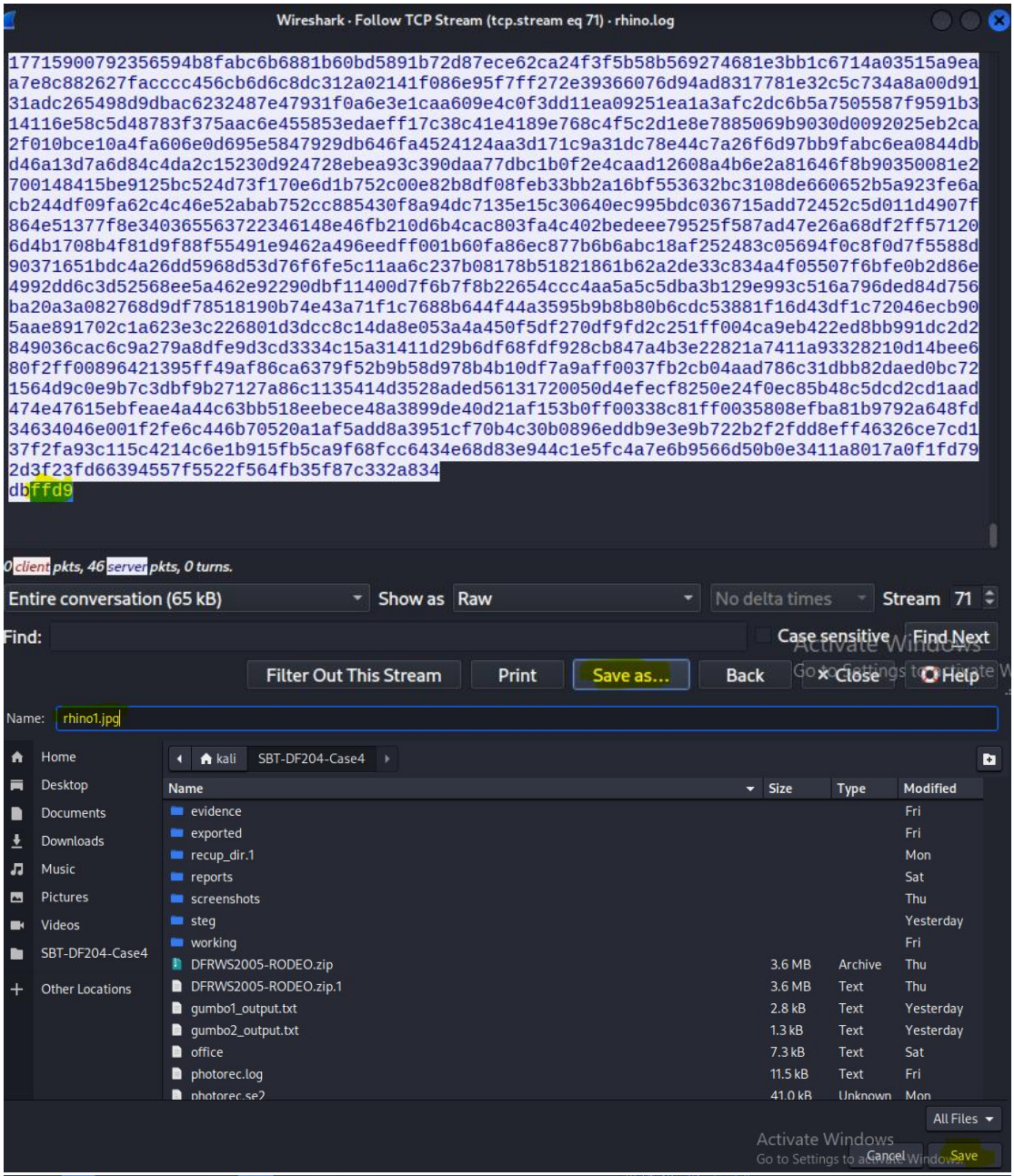
Item	Findings
	V1.01”
Steganography detector output	stegdetect flagged f0105065.jpg and f0104249.jpg positive. stegbreak -f rockyou.txt confirmed both as **JPHide v5**-encoded, cracking passphrases via dictionary attack: f0105065.jpg >> passphrase gator; f0104249.jpg >> passphrase gumbo.
Notable document	f0335017_...doc, “She died in February at the age of 74,” created 2005-08-09 03:17, last saved 2005-08-09 03:40, author field reads “NO WAY MAN NO WAY MAN NOWAY”
Evidence that supports a hiding interpretation	**Confirmed steganography.** Both files contained JPHide-concealed data, successfully extracted using jpseek.exe (via Wine) with the cracked passphrases. Extracted outputs r065.jpg and r249.jpg are both valid, viewable JPEG images (confirmed with display), which means real hidden image content was successfully recovered from within the carrier JPEGs.
Authorised recovery workflow used	stegdetect/stegbreak (built from source, Stegdetect 0.6.1) for detection and passphrase cracking; jpseek.exe (JPHide/JPSeek suite v0.51, via Wine) for extraction using the recovered passphrases
Extracted content	r065.jpg, extracted from f0105065.jpg using passphrase gator; r249.jpg, extracted from f0104249.jpg using passphrase gumbo
Recovered file(s), type and hash	r065.jpg - JPEG, 52 kB, MD5 63a39823f80b321c2dcd112158b55011, SHA-256 bc9c605372fdf9cc00faf17ae9ce0fdb6d26bb9db28cdff742d9ddbe1470ec16. r249.jpg - JPEG, 41 kB, MD5 87018ef0cfdb91e818d92efeb9c19338, SHA-256 6dbf775b8999a006ffa74c65127e5add64234184bd2f425b2ad644e8c0d19345
Metadata review findings (exiftool)	r065.jpg: captured on a **Nikon Coolpix E950**, original capture date **19 Mar 2001, 13:11:20**, 400×340 resolution, later processed with **ACD Systems Digital Imaging** software with a modify date of **8 Aug 2005**. Contains an embedded thumbnail. r249.jpg: 384×256 resolution, also processed with ACD Systems Digital Imaging, modify date **8 Aug 2005**, no camera make/model or original capture date recorded (stripped or never present)

7. Part C - Reconstruct and Validate FTP Evidence




1 2 3 4
Main Toolbar
Filter Toolbar
Wireless Toolbar
Status Bar
Full Screen F11
Packet List
Packet Details
Packet Bytes
Packet Diagram
Time Display Format
Name Resolution
Zoom
Expand Subtrees Shift+Right
Collapse Subtrees Shift+Left
Expand All Ctrl+Right
Collapse All Ctrl+Left
Colorize Packet List
Coloring Rules…
Colorize Conversation
Reset Layout Ctrl+Shift+W
Resize Columns Ctrl+Shift+R
Date and Time of Day (1970-01-01 01:02:03.123456) Ctrl+Alt+1
Year, Day of Year, and Time of Day (1970/001 01:02:03.123456)
Time of Day (01:02:03.123456) Ctrl+Alt+2
Seconds Since 1970-01-01 Ctrl+Alt+3
Seconds Since First Captured Packet Ctrl+Alt+4
Seconds Since Previous Captured Packet Ctrl+Alt+5
Seconds Since Previous Displayed Packet Ctrl+Alt+6
UTC Date and Time of Day (1970-01-01 01:02:03.123456Z) Ctrl+Alt+7
UTC Year, Day of Year, and Time of Day (1970/001 01:02:03.123456Z)
UTC Time of Day (01:02:03.123456Z) Ctrl+Alt+8
Automatic (from capture file)
Seconds
Tenths of a second
Hundredths of a second
Milliseconds
Tenths of a millisecond
Hundredths of a millisecond
Microseconds
Tenths of a microsecond
Hundredths of a microsecond
Nanoseconds
Display Seconds With Hours and Minutes
Activate Window
Go to Settings to acti

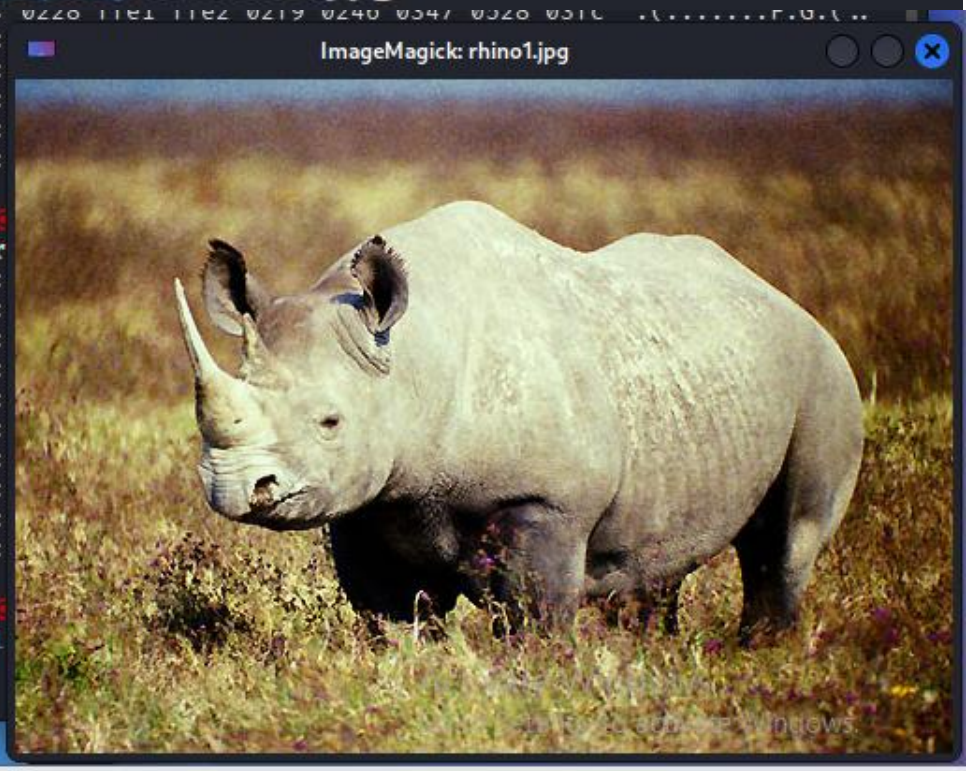

Wireshark · Follow TCP Stream (tcp.stream eq 71) · rhino.log
ffd8ffe000104a46494600010201004800480000ffed159850686f746f73686f7020332e30003842494d03e900
000000007800030000004800480000000002d80228ffe1ffe202f902460347052803fc0002000004800480000
000002d80228000100000064000000010003030300000001270f0001000100000000000000000000000600800
190190000000000000000000000000000000000000000000000000000000000000000000000003842494d03ed00000000
001000480000000100010048000000010001 3842494d03f3000000000080000000000000003842494d040a00
000000001000003842494d271000000000000a0001000000000000000023842494d03f5000000000048002f6666
0001006c6666000600000000001002f6666000100a1999a0006000000000001003200000001005a0000000600
000000001003500000001002d000000600000000000013842494d03f8000000000700000ffffffffffffffff
ffffffffffffffffffffffffffff03e800000000ffffffffffffffffffffffffffffffffffffffffffffffff03e800
000000ffffffffffffffffffffffffffffffffffffffffffffff03e800000000ffffffffffffffffffffffffffffffff
ffffffffffffffff03e800003842494d040800000000000100000000100000240000002400000000003842494d04
090000000013a3000000010000008000000005b000001800000888000001387001800001ffd8ffe000104a464946
000102010048004800000ffe002746696c65207772697474656e2062792041646f626520050686f746f73686f70
a820342e3000ffee000e41646f626500648000000001ffdb0084000c08080809080c09090c110b0a0b11150f0c
0c0f151813131513131811 0c0c0c0c0c0c110c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c
0c010d0b0b0d0e0d100e0e10140e0e0e14140e0e0e0e14110c0c0c0c0c11110c0c0c0c0c0c110c0c0c0c0c0c0c
0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0c0cffc0001108005b008003012200021101031101ffdd000400
08ffc4013f000001050101010101010000000000000003000102040506070809 0a0b0100010501010101010100
000000000000010002030405060708090a0b1000010401030204020507060805030c330100021103042112310 5
4151611322718132061491a1b14223241552c16233347282d14307259253f0e1f163733516a2b2832644935464
45c2a3743617d255e265f2b384c3d375e3f3462794a485b495c4d4e4f4a5b5c5d5e5f55666768696a6b6c6d6e6
f637475767778797a7b7c7d7e7f711000202010204040304050607070605350100021103213112044151617122
130532819114a1b14223c152d1f0332462e17282924353156373 34f1250616a2b283072635c2d2449354a31764
4555367465e2f2b384c3d375e3f34694a485b495c4d4e4f4a5b5c5d5e5f55666768696a6b6c6d6e6f627374757

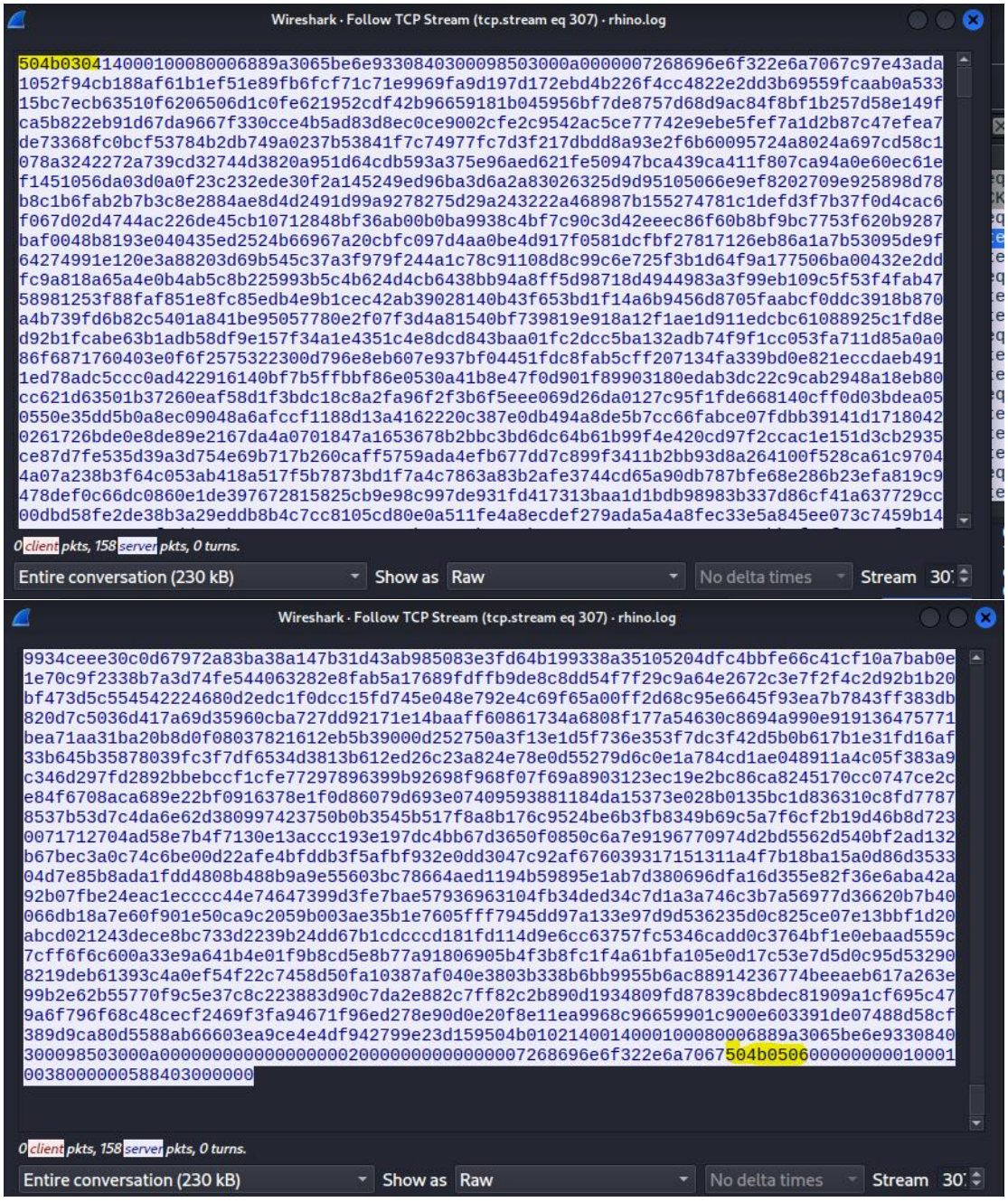

Wireshark · Follow TCP Stream (tcp.stream eq 71) · rhino.log
17715900792356594b8fabc6b6881b60bd5891b72d87ece62ca24f3f5b58b569274681e3bb1c6714a03515a9ea
a7e8c882627facccc456cb6d6c8dc312a02141f086e95f7ff272e39366076d94ad8317781e32c5c734a8a00d91
31adc265498d9dbac6232487e47931f0a6e3e1caa609e4c0f3dd11ea09251ea1a3afc2dc6b5a7505587f9591b3
14116e58c5d48783f375aac6e455853edaeff17c38c41e4189e768c4f5c2d1e8e7885069b9030d0092025eb2ca
2f010bce10a4fa606e0d695e5847929db646fa4524124aa3d171c9a31dc78e44c7a26f6d97bb9fabc6ea0844db
d46a13d7a6d84c4da2c15230d924728ebea93c390daa77dbc1b0f2e4caad12608a4b6e2a81646f8b90350081e2
700148415be9125bc524d73f170e6d1b752c00e82b8df08feb33bb2a16bf553632bc3108de660652b5a923fe6a
cb244df09fa62c4c46e52abab752cc885430f8a94dc7135e15c30640ec995bdc036715add72452c5d011d4907f
864e51377f8e340365563722346148e46fb210d6b4cac803fa4c402bedeee79525f587ad47e26a68df2ff57120
6d4b1708b4f81d9f88f55491e9462a496eedff001b60fa86ec877b6b6abc18af252483c05694f0c8f0d7f5588d
90371651bdc4a26dd5968d53d76f6fe5c11aa6c237b08178b51821861b62a2de33c834a4f05507f6bfe0b2d86e
4992dd6c3d52568ee5a462e92290dbf11400d7f6b7f8b22654ccc4aa5a5c5dba3b129e993c516a796ded84d756
ba20a3a082768d9df78518190b74e43a71f1c7688b644f44a3595b9b8b80b6cdc53881f16d43df1c72046ecb90
5aae891702c1a623e3c226801d3dcc8c14da8e053a4a450f5df270df9fd2c251ff004ca9eb422ed8bb991dc2d2
849036cac6c9a279a8dfe9d3cd3334c15a31411d29b6df68fdf928cb847a4b3e22821a7411a93328210d14bee6
80f2ff00896421395ff49af86ca6379f52b9b58d978b4b10df7a9aff0037fb2cb04aad786c31dbb82daed0bc72
1564d9c0e9b7c3dbf9b27127a86c1135414d3528aded56131720050d4efecf8250e24f0ec85b48c5dcd2cd1aad
474e47615ebfeae4a44c63bb518eebece48a3899de40d21af153b0ff00338c81ff0035808efba81b9792a648fd
34634046e001f2fe6c446b70520a1af5add8a3951cf70b4c30b0896eddb9e3e9b722b2f2fdd8eff46326ce7cd1
37f2fa93c115c4214c6e1b915fb5ca9f68fcc6434e68d83e944c1e5fc4a7e6b9566d50b0e3411a8017a0f1fd79
2d3f23fd66394557f5522f564fb35f87c332a834
dbffd9
0 client pkts, 46 server pkts, 0 turns.
Entire conversation (65 kB)
Show as
Raw
No delta times
Stream
71
Find:
Case sensitive
Find Next
Filter Out This Stream
Print
Save as...
Back
Close
Help
Name:
rhino1.jpg
Home
Desktop
Documents
Downloads
Music
Pictures
Videos
SBT-DF204-Case4
Other Locations
kali
SBT-DF204-Case4
Name
Size
Type
Modified
evidence
exported
recup_dir.1
reports
screenshots
steg
working
DFRWS2005-RODEO.zip
DFRWS2005-RODEO.zip.1
gumbo1_output.txt
gumbo2_output.txt
office
photorec.log
photorec.se2
3.6 MB
3.6 MB
2.8 kB
1.3 kB
7.3 kB
11.5 kB
41.0 kB
Archive
Text
Text
Text
Text
Unknown
Fri
Fri
Mon
Sat
Thu
Yesterday
Fri
Thu
Thu
Yesterday
Yesterday
Sat
Fri
Mon
All Files
Cancel
Save
Activate Windows
Go to Settings to activate Windows.
rhino.log
File
Edit
View
Go
Capture
Analyze
Statistics
Telephony
Wireless
Tools
Help
Apply a display filter … <Ctrl-/>
No. Time Source Destination Protocol Length Info
1542 2004-04-26 22:21:46.668224Z 137.30.122.253 137.30.120.40 TCP 54 1655 → 21
1543 2004-04-26 22:21:46.688481Z 128.138.182.127 137.30.123.164 TCP 62 [TCP Retra
1544 2004-04-26 22:21:49.952880Z 137.30.122.253 137.30.120.40 FTP 81 Request: P
1545 2004-04-26 22:21:49.953485Z 137.30.120.40 137.30.122.253 FTP 84 Response:
1546 2004-04-26 22:21:49.954047Z 137.30.122.253 137.30.120.40 FTP 71 Request: S
1547 2004-04-26 22:21:49.984631Z 137.30.120.40 137.30.122.253 TCP 62 20 → 1657
1548 2004-04-26 22:21:49.985023Z 137.30.122.253 137.30.120.40 TCP 62 1657 → 20
1549 2004-04-26 22:21:49.985342Z 137.30.120.40 137.30.122.253 TCP 60 20 → 1657
1550 2004-04-26 22:21:49.991431Z 137.30.120.40 137.30.122.253 FTP 111 Response:
1551 2004-04-26 22:21:49.995482Z 137.30.122.253 137.30.120.40 FTP-DA… 1514 FTP Data:
1552 2004-04-26 22:21:49.995629Z 137.30.122.253 137.30.120.40 FTP-DA… 1514 FTP Data:
1553 2004-04-26 22:21:49.996045Z 137.30.120.40 137.30.122.253 TCP 60 20 → 1657
1554 2004-04-26 22:21:49.996157Z 137.30.122.253 137.30.120.40 FTP-DA… 1514 FTP Data:
1555 2004-04-26 22:21:49.996234Z 137.30.122.253 137.30.120.40 FTP-DA… 1514 FTP Data:
1556 2004-04-26 22:21:49.996246Z 137.30.120.40 137.30.122.253 TCP 60 20 → 1657
1557 2004-04-26 22:21:49.996334Z 137.30.122.253 137.30.120.40 FTP-DA… 1514 FTP Data:
1558 2004-04-26 22:21:49.996413Z 137.30.122.253 137.30.120.40 FTP-DA… 1514 FTP Data:

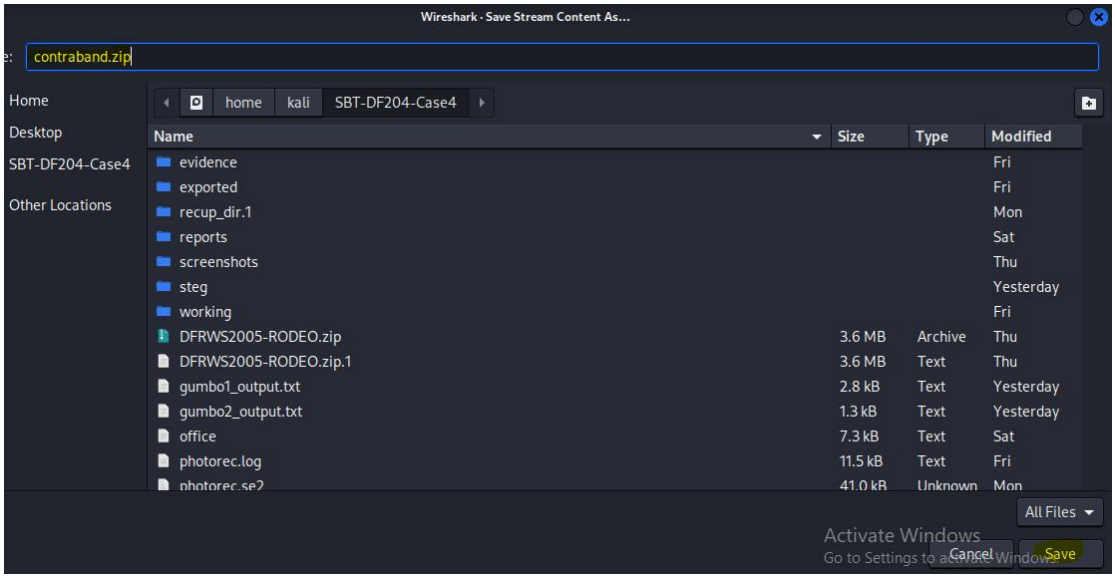
```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# xxd rhino1.jpg | head
00000000: ffd8 ffe0 0010 4a46 4946 0001 0201 0048  ......JFIF.....H
00000010: 0048 0000 ffed 1598 5068 6f74 6f73 686f  .H......Photosho
00000020: 7020 332e 3000 3842 494d 03e9 0000 0000  p 3.0.8BIM......
00000030: 0078 0003 0000 0048 0048 0000 0000 02d8  .x.....H.H......
00000040: 0228 ffe1 ffe2 02f9 0246 0347 0528 03fc  .(.......F.G.(..
00000050: 0002 0000 0048 0048 0000 0000 02d8 0228  .....H.H.......(
00000060: 0001 0000 0064 0000 0001 0003 0303 0000  .....d..........
00000070: 0001 270f 0001 0001 0000 0000 0000 0000  ..'.............
00000080: 0000 0000 6008 0019 0190 0000 0000 0000  ....`...........
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  ................

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# xxd rhino1.jpg | tail
00010010: e4a4 4c63 bb51 8eeb ece4 8a38 99de 40d2  ..Lc.Q.....8..@.
00010020: 1af1 53b0 ff00 338c 81ff 0035 808e fba8  ..S...3....5....
00010030: 1b97 92a6 48fd 3463 4046 e001 f2fe 6c44  ....H.4c@F....lD
00010040: 6b70 520a 1af5 add8 a395 1cf7 0b4c 30b0  kpR..........L0.
00010050: 896e ddb9 e3e9 b722 b2f2 fdd8 eff4 6326  .n.....".......c&
00010060: ce7c d137 f2fa 93c1 15c4 214c 6e1b 915f  .|.7......!Ln.._
00010070: b5ca 9f68 fcc6 434e 68d8 3e94 4c1e 5fc4  ...h..CNh.>.L._.
00010080: a7e6 b956 6d50 b0e3 411a 8017 a0f1 fd79  ...VmP..A......y
00010090: 2d3f 23fd 6639 4557 f552 2f56 4fb3 5f87  -?#.f9EW.R/VO._.
000100a0: c332 a834 dbff d9                        .2.4...
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# display rhino1.jpg
```




Wireshark · Follow TCP Stream (tcp.stream eq 307) · rhino.log
504b0304140001000800068 89a3065be6e9330840300098503000a0000007268696e6f322e6a7067c97e43ada
1052f94cb188af61b1ef51e89fb6fcf71c71e9969fa9d197d172ebd4b226f4cc4822e2dd3b69559fcaab0a533
15bc7ecb63510f6206506d1c0fe621952cdf42b96659181b045956bf7de8757d68d9ac84f8bf1b257d58e149f
ca5b822eb91d67da9667f330cce4b5ad83d8ec0ce9002cfe2c9542ac5ce77742e9ebe5fef7a1d2b87c47efea7
de73368fc0bcf53784b2db749a0237b53841f7c74977fc7d3f217dbdd8a93e2f6b60095724a8024a697cd58c1
078a3242272a739cd32744d3820a951d64cdb593a375e96aed621fe50947bca439ca411f807ca94a0e60ec61e
f1451056da03d0a0f23c232ede30f2a145249ed96ba3d6a2a83026325d9d95105066e9ef8202709e925898d78
b8c1b6fab2b7b3c8e2884ae8d4d2491d99a9278275d29a243222a468987b155274781c1defd3f7b37f0d4cac6
f067d02d4744ac226de45cb10712848bf36ab00b0ba9938c4bf7c90c3d42eeec86f60b8bf9bc7753f620b9287
baf0048b8193e040435ed2524b66967a20cbfc097d4aa0be4d917f0581dcfbf27817126eb86a1a7b53095de9f
64274991e120e3a88203d69b545c37a3f979f244a1c78c91108d8c99c6e725f3b1d64f9a177506ba00432e2dd
fc9a818a65a4e0b4ab5c8b225993b5c4b624d4cb6438bb94a8ff5d98718d4944983a3f99eb109c5f53f4fab47
58981253f88faf851e8fc85edb4e9b1cec42ab39028140b43f653bd1f14a6b9456d8705faabcf0ddc3918b870
a4b739fd6b82c5401a841be95057780e2f07f3d4a81540bf739819e918a12f1ae1d911edcbc61088925c1fd8e
d92b1fcabe63b1adb58df9e157f34a1e4351c4e8dcd843baa01fc2dcc5ba132adb74f9f1cc053fa711d85a0a0
86f6871760403e0f6f2575322300d796e8eb607e937bf04451fdc8fab5cff207134fa339bd0e821eccdaeb491
1ed78adc5ccc0ad422916140bf7b5ffbbf86e0530a41b8e47f0d901f89903180edab3dc22c9cab2948a18eb80
cc621d63501b37260eaf58d1f3bdc18c8a2fa96f2f3b6f5eee069d26da0127c95f1fde668140cff0d03bdea05
0550e35dd5b0a8ec09048a6afccf1188d13a4162220c387e0db494a8de5b7cc66fabce07fdbb39141d1718042
0261726bde0e8de89e2167da4a0701847a1653678b2bbc3bd6dc64b61b99f4e420cd97f2ccac1e151d3cb2935
ce87d7fe535d39a3d754e69b717b260caff5759ada4efb677dd7c899f3411b2bb93d8a264100f528ca61c9704
4a07a238b3f64c053ab418a517f5b7873bd1f7a4c7863a83b2afe3744cd65a90db787bfe68e286b23efa819c9
478def0c66dc0860e1de397672815825cb9e98c997de931fd417313baa1d1bdb98983b337d86cf41a637729cc
00dbd58fe2de38b3a29eddb8b4c7cc8105cd80e0a511fe4a8ecdef279ada5a4a8fec33e5a845ee073c7459b14
0 client pkts, 158 server pkts, 0 turns.
Entire conversation (230 kB)
Show as
Raw
No delta times
Stream 30
Wireshark · Follow TCP Stream (tcp.stream eq 307) · rhino.log
9934ceee30c0d67972a83ba38a147b31d43ab985083e3fd64b199338a35105204dfc4bbfe66c41cf10a7bab0e
1e70c9f2338b7a3d74fe544063282e8fab5a17689fdffb9de8c8dd54f7f29c9a64e2672c3e7f2f4c2d92b1b20
bf473d5c554542224680d2edc1f0dcc15fd745e048e792e4c69f65a00ff2d68c95e6645f93ea7b7843ff383db
820d7c5036d417a69d35960cba727dd92171e14baaff60861734a6808f177a54630c8694a990e919136475771
bea71aa31ba20b8d0f08037821612eb5b39000d252750a3f13e1d5f736e353f7dc3f42d5b0b617b1e31fd16af
33b645b35878039fc3f7df6534d3813b612ed26c23a824e78e0d55279d6c0e1a784cd1ae048911a4c05f383a9
c346d297fd2892bbebccf1cfe77297896399b92698f968f07f69a8903123ec19e2bc86ca8245170cc0747ce2c
e84f6708aca689e22bf0916378e1f0d86079d693e0740959388 1184da15373e028b0135bc1d836310c8fd7787
8537b53d7c4da6e62d380997423750b0b3545b517f8a8b176c9524be6b3fb8349b69c5a7f6cf2b19d46b8d723
0071712704ad58e7b4f7130e13accc193e197dc4bb67d3650f0850c6a7e9196770974d2bd5562d540bf2ad132
b67bec3a0c74c6be00d22afe4bfddb3f5afbf932e0dd3047c92af676039317151311a4f7b18ba15a0d86d3533
04d7e85b8ada1fdd4808b488b9a9e55603bc78664aed1194b59895e1ab7d380696dfa16d355e82f36e6aba42a
92b07fbe24eac1ecccc44e74647399d3fe7bae57936963104fb34ded34c7d1a3a746c3b7a56977d36620b7b40
066db18a7e60f901e50ca9c2059b003ae35b1e7605fff7945dd97a133e97d9d536235d0c825ce07e13bbf1d20
abcd021243dece8bc733d2239b24dd67b1cdcccd181fd114d9e6cc63757fc5346cadd0c3764bf1e0ebaad559c
7cff6f6c600a33e9a641b4e01f9b8cd5e8b77a91806905b4f3b8fc1f4a61bfa105e0d17c53e7d5d0c95d53290
8219deb61393c4a0ef54f22c7458d50fa10387af040e3803b338b6bb9955b6ac88914236774beeaeb617a263e
99b2e62b55770f9c5e37c8c223883d90c7da2e882c7ff82c2b890d1934809fd87839c8bdec81909a1cf695c47
9a6f796f68c48cecf2469f3fa94671f96ed278e90d0e20f8e11ea9968c96659901c900e603391de07488d58cf
389d9ca80d5588ab66603ea9ce4e4df942799e23d159504b0102140014000100080006889a3065be6e9330840
300098503000a0000000000000000002000000000000007268696e6f322e6a7067504b0506000000000100010
0038000000588403000000
0 client pkts, 158 server pkts, 0 turns.
Entire conversation (230 kB)
Show as
Raw
No delta times
Stream 30




```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# ls -l contraband.zip
-rw-r--r-- 1 root root 230566 Oct 28 18:52 contraband.zip
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# xxd contraband.zip | head
00000000: 504b 0304 1400 0100 0800 0688 9a30 65be  PK...........0e.
00000010: 6e93 3084 0300 0985 0300 0a00 0000 7268  n.0...........rh
00000020: 696e 6f32 2e6a 7067 c97e 43ad a105 2f94  ino2.jpg.~C .../.
00000030: cb18 8af6 1b1e f51e 89fb 6fcf 71c7 1e99  ..........o.q...
00000040: 69fa 9d19 7d17 2ebd 4b22 6f4c c482 2e2d  i...}...K"oL...-
00000050: d3b6 9559 fcaa b0a5 3315 bc7e cb63 510f  ...Y....3..~.cQ.
00000060: 6206 506d 1c0f e621 952c df42 b966 5918  b.Pm...!.,.B.fY.
00000070: 1b04 5956 bf7d e875 7d68 d9ac 84f8 bf1b  ..YV.}.u}h......
00000080: 257d 58e1 49fc a5b8 22eb 91d6 7da9 667f  %}X.I...".....}.f.
00000090: 330c ce4b 5ad8 3d8e c0ce 9002 cfe2 c954  3..KZ.=........T

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# xxd contraband.zip | tail
00038410: 9a1c f695 c479 a6f7 96f6 8c48 cecf 2469  .....y.....H..$i
00038420: f3fa 9467 1f96 ed27 8e90 d0e2 0f8e 11ea  ...g...'........
00038430: 9968 c966 5990 1c90 0e60 3391 de07 488d  .h.fY....`3...H.
00038440: 58cf 389d 9ca8 0d55 88ab 6660 3ea9 ce4e  X.8....U..f`>..N
00038450: 4df9 4279 9e23 d159 504b 0102 1400 1400  M.By.#.YPK......
00038460: 0100 0800 0688 9a30 65be 6e93 3084 0300  .......0e.n.0...
00038470: 0985 0300 0a00 0000 0000 0000 0000 2000  .............. .
00038480: 0000 0000 0000 7268 696e 6f32 2e6a 7067  ......rhino2.jpg
00038490: 504b 0506 0000 0000 0100 0100 3800 0000  PK..........8...
000384a0: 5884 0300 0000                           X.....
```

Item	Findings
Control-session stream and endpoints	TCP stream (tcp.stream eq 69), FTP control channel on port 21. Server source: 137.30.122.253, destination port 1655. Full

Item	Findings
	conversation length: 487 bytes (6 client packets, 9 server packets, 12 turns)
Login identifier observed (redact credential in screenshots)	Username: gnome - confirmed via Wireshark's built-in **Credentials** panel (Tools, Credentials), which flagged FTP usernames in packets **1536, 1629, and 5637**, each logged as "Username in packet: [packet #]"
Transfer command and target filename	TYPE I (binary mode set) >> PORT 137,30,122,253,6,121 (active-mode data connection negotiated) >> STOR rhino1.jpg - Server response: 150 Opening BINARY mode data connection for rhino1.jpg. followed by 226 Transfer complete. Direction: **client >> server (upload)**, not a download. Total of **65703 bytes transferred in 1 file** client uploaded a file named rhino1.jpg to the server
Data-session stream, direction, packet range	Server response: 150 Opening BINARY mode data connection for rhino1.jpg. followed by 226 Transfer complete. Direction: **client >> server (upload)**, not a download. Total of **65703 bytes transferred in 1 file**
Server response codes observed	331 (password required), 230 (user logged in), 200 (command successful, x2, for TYPE and PORT), 150 (opening data connection), 226 (transfer complete), 221 (session closing: "You have transferred 65703 bytes in 1 files")
Reconstructed object: type, size, hash	rhino1.jpg, uploaded via this FTP session, **65,703 bytes**
Matches an artefact recovered elsewhere? (compare hashes)	**Yes,** the 65,703-byte size matches exactly the rhino1.jpg already present in your case working directory (confirmed earlier: -rw-rw-r-- 1 kali kali 65703 Oct 28 17:33 rhino1.jpg). This strongly suggests the rhino1.jpg file you've been analyzing throughout this case *is* the file captured being transferred in this FTP session
Password / authentication	PASS gnome123: server responded 230 User gnome logged in. confirming successful authentication

8. Part D - The Protected Archive

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# unzip contraband.zip
Archive:  contraband.zip
[contraband.zip] rhino2.jpg password:
```

```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# apt-get install fcrackzip
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
fcrackzip is already the newest version (1.0-12).
Solving dependencies ... Done
The following package was automatically installed and is no longer required:
  python3.13-tk
Use 'sudo apt autoremove' to remove it.
0 upgraded, 0 newly installed, 0 to remove and 597 not upgraded.
```

```
┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/USB_image/DFRWS2005-RODEO.zip

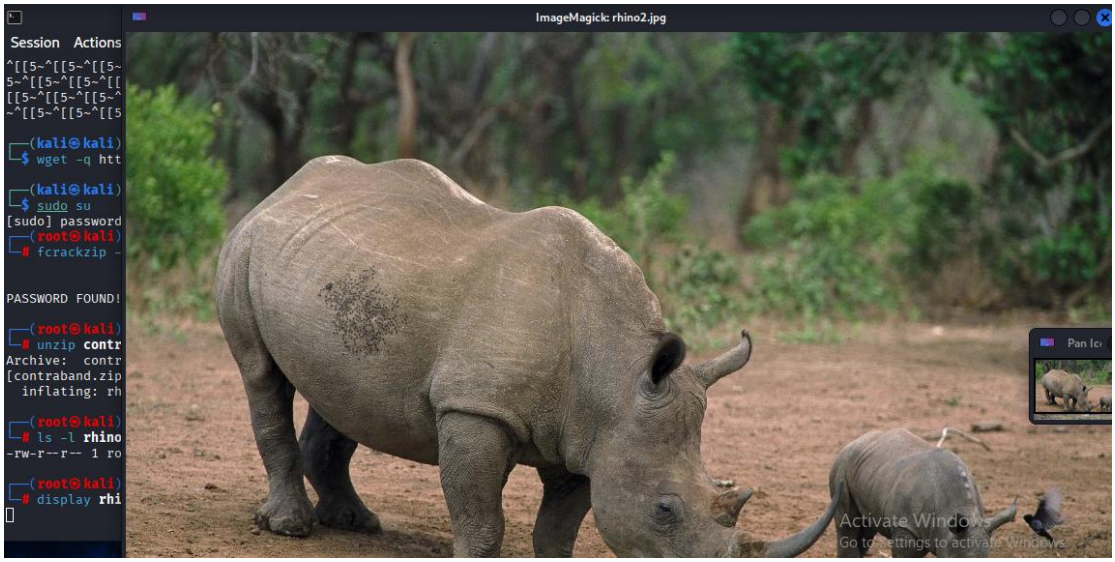
┌──(kali㉿kali)-[~/SBT-DF204-Case4]
└─$ sudo su
[sudo] password for kali:
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# fcrackzip -u -D -p rockyou.txt contraband.zip

PASSWORD FOUND!!!!: pw == monkey

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# unzip contraband.zip
Archive:  contraband.zip
[contraband.zip] rhino2.jpg password:
  inflating: rhino2.jpg

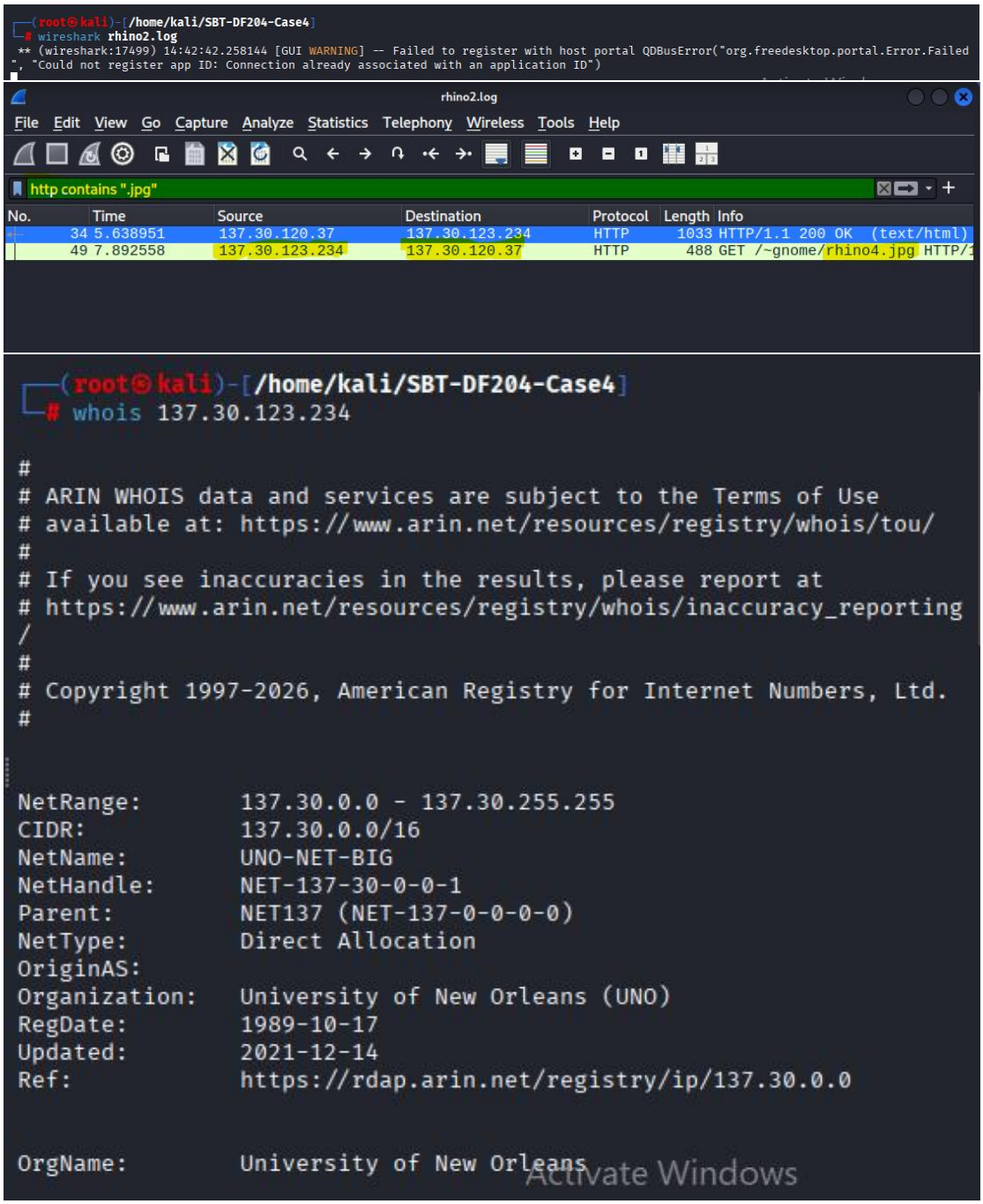
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# ls -l rhino2.jpg
-rw-r--r-- 1 root root 230665 Apr 26  2004 rhino2.jpg

┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# display rhino2.jpg
```



Item	Your Finding
Archive completeness (unzip -l check)	contraband.zip identified as containing a single compressed member, rhino2.jpg. Archive is password-protected (standard ZIP encryption), unzip prompts interactively for a password before extraction will proceed
Evidence the archive was encrypted	Attempting direct extraction returns: [contraband.zip] rhino2.jpg password: confirming password,protected/encrypted archive rather than a plain compressed file
Authorised recovery workflow used (course tool/method only)	fcrackzip dictionary (brute-force) attack using the rockyou.txt wordlist: fcrackzip -u -D -p rockyou.txt contraband.zip. Password recovered: monkey
Extracted content: type, size, hash	rhino2.jpg (JPEG), **230,665 bytes**, extracted successfully using the recovered password. Visually confirmed viewable (opened with display rhino2.jpg without error)
Matches another artefact in the case? (compare hashes)	**Yes, it matches,** 230,665 bytes is the exact same file size independently recovered by PhotoRec carving for f0106409.jpg

8. Part E - Reconstruct and Validate HTTP Evidence



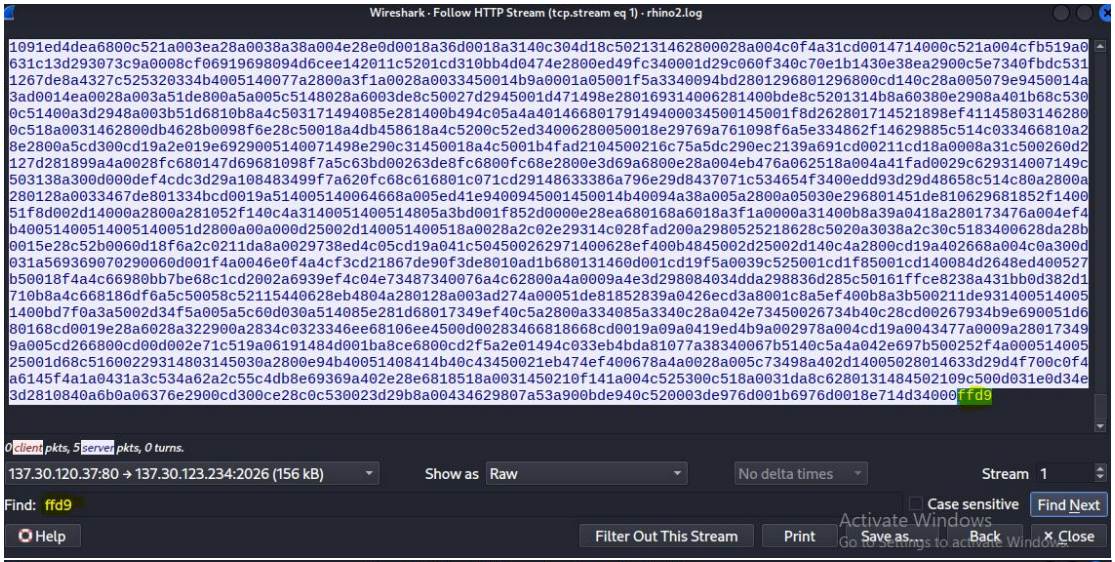
```
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

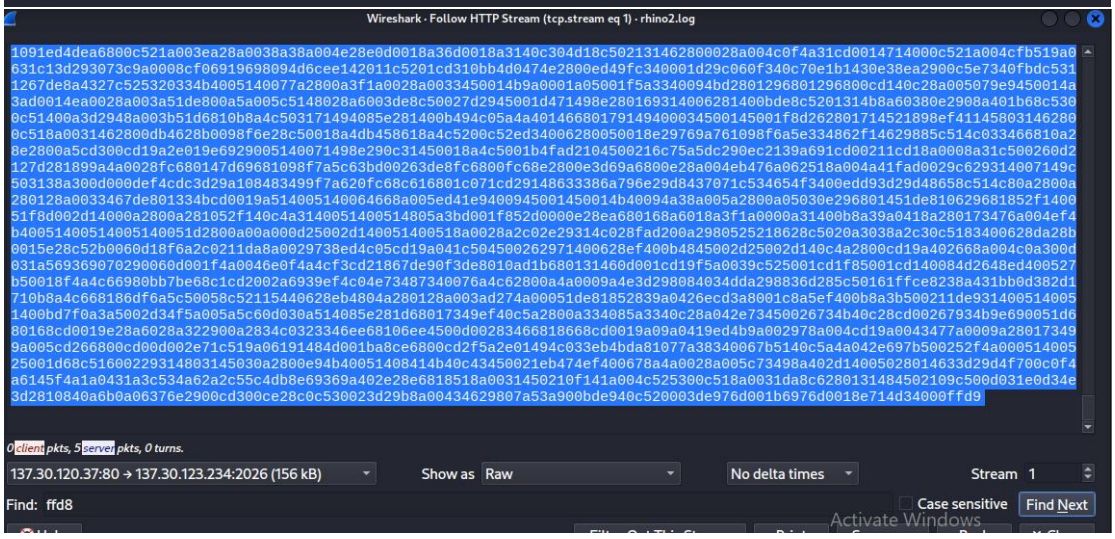

NetRange:       137.30.0.0 - 137.30.255.255
CIDR:           137.30.0.0/16
NetName:        UNO-NET-BIG
NetHandle:      NET-137-30-0-0-1
Parent:         NET137 (NET-137-0-0-0-0)
NetType:        Direct Allocation
OriginAS:
Organization:   University of New Orleans (UNO)
RegDate:        1989-10-17
Updated:        2021-12-14
Ref:            https://rdap.arin.net/registry/ip/137.30.0.0

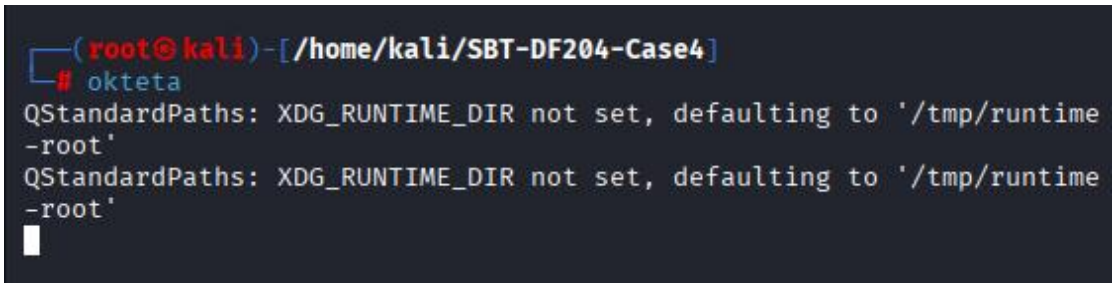

OrgName:        University of New Orleans
OrgId:          UNO
Address:        Computer Research Building
Address:         Lakefront Campus
City:           New Orleans
StateProv:      LA
PostalCode:     70148
Country:        US
RegDate:        1988-08-17
Updated:        2015-10-23
Ref:            https://rdap.arin.net/registry/entity/UNO
```

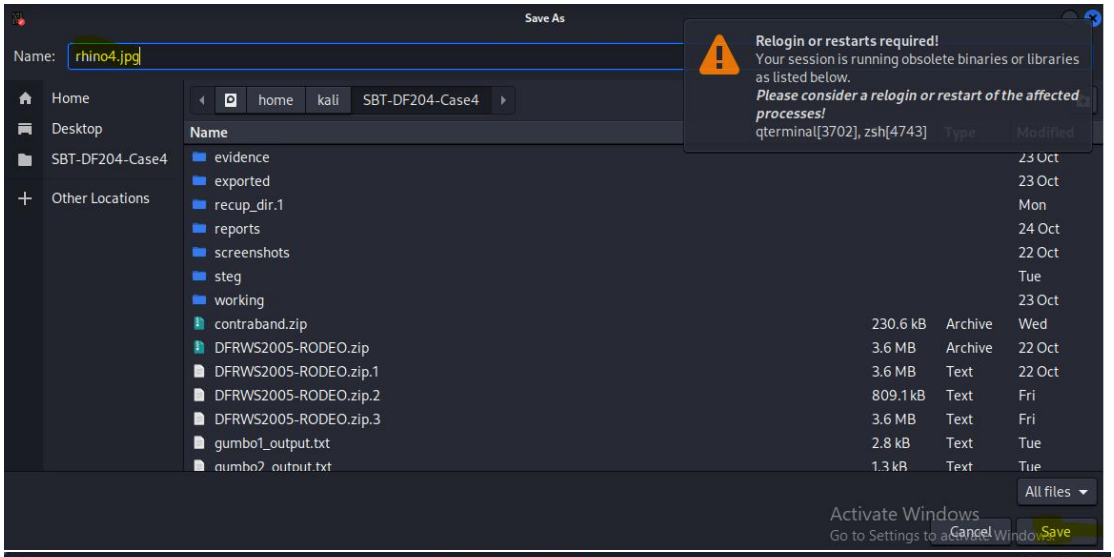
whois 137.30.123.234

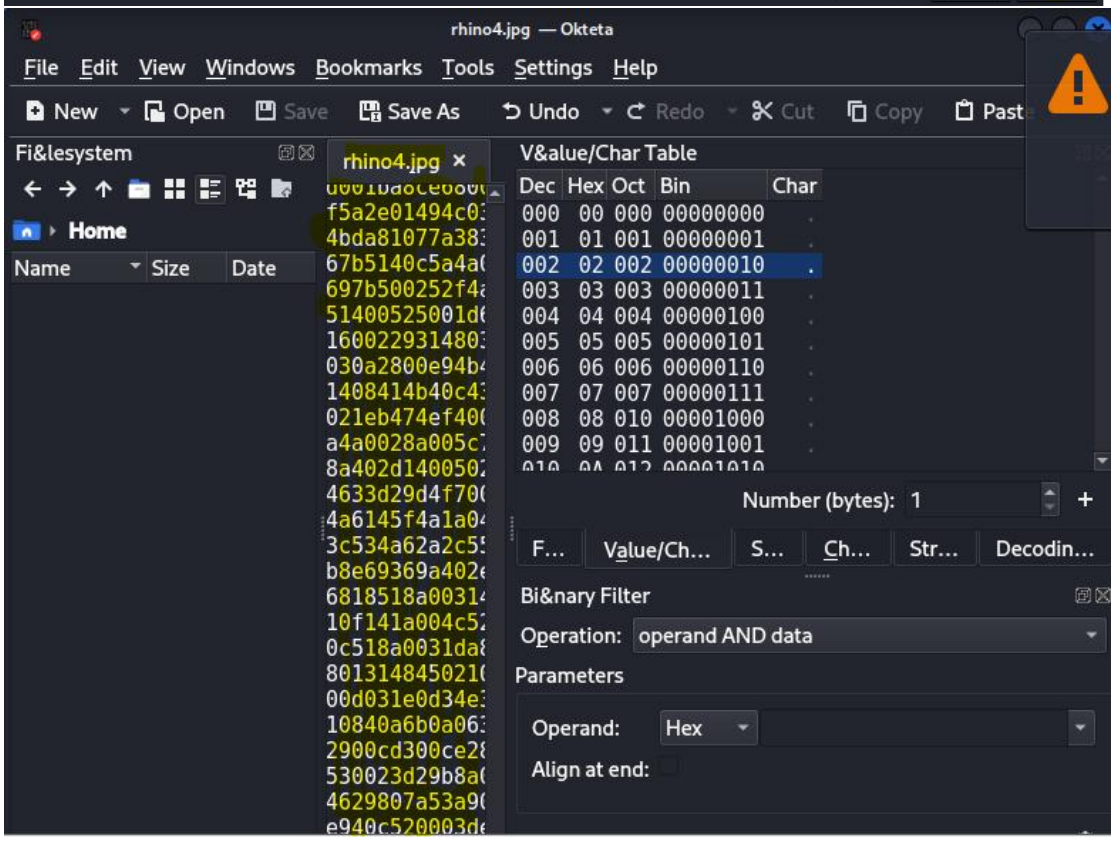



Wireshark · Follow HTTP Stream (tcp.stream eq 1) · rhino2.log
1091ed4dea6800c521a003ea28a0038a38a004e28e0d0018a36d0018a3140c304d18c502131462800028a004c0f4a31cd0014714000c521a004cfb519a0
631c13d293073c9a0008cf06919698094d6cee142011c5201cd310bb4d0474e2800ed49fc340001d29c060f340c70e1b1430e38ea2900c5e7340fbdc531
1267de8a4327c525320334b4005140077a2800a3f1a0028a0033450014b9a0001a05001f5a3340094bd2801296801296800cd140c28a005079e9450014a
3ad0014ea0028a003a51de800a5a005c5148028a6003de8c50027d2945001d471498e2801693140062814 00bde8c5201314b8a60380e2908a401b68c530
0c51400a3d2948a003b51d6810b8a4c503171494085e281400b494c05a4a401466801791494000345001450 01f8d2628017145218 98ef41145803146280
0c518a0031462800db4628b0098f6e28c50018a4db458618a4c5200c52ed34006280050018e29769a761098f6a5e334862f14629885c514c033466810a2
8e2800a5cd300cd19a2e019e69290051400714 98e290c31450018a4c5001b4fad2104500216c75a5dc290ec2139a691cd00211cd18a0008a31c500260d2
127d281899a4a0028fc680147d69681098f7a5c63bd00263de8fc6800fc68e2800e3d69a6800e28a004eb476a062518a004a41fad0029c629314007149c
503138a300d000def4cdc3d29a108483499f7a620fc68c616801c071cd29148633386a796e29d8437071c534654f3400edd93d29d48658c514c80a2800a
280128a0033467de801334bcd0019a514005140064668a005ed41e940094500145001 4b40094a38a005a2800a05030e296801451de810629681852f1400
51f8d002d14000a2800a281052f140c4a3140051400514805a3bd001f852d0000e28ea680168a6018a3f1a0000a31400b8a39a0418a280173476a004ef4
b400514005140051400514051d2800a00a000d25002d140051400518a0028a2c02e29314c028fad200a2980525218628c5020a3038a2c30c5183400628da28b
0015e28c52b0060d18f6a2c0211da8a0029738ed4c05cd19a041c5045002629714006 28ef400b4845002d25002d140c4a2800cd19a402668a004c0a300d
031a569369070290060d001f4a0046e0f4a4cf3cd21867de90f3de8010ad1b680131460d001cd19f5a0039c525001cd1f85001cd140084d2648ed400527
b50018f4a4c66980bb7be68c1cd2002a6939ef4c04e73487340076a4c62800a4a0009a4e3d298084034dda298836d285c50161ffce8238a431bb0d382d1
710b8a4c668186df6a5c50058c52115440628eb4804a280128a003ad274a00051de81852839a0426ecd3a8001c8a5ef400b8a3b500211de931400514005
1400bd7f0a3a5002d34f5a005a5c60d030a514085e281d68017349ef40c5a2800a334085a3340c28a042e73450026734b40c28cd00267934b9e690051d6
80168cd0019e28a6028a322900a2834c0323346ee68106ee4500d00283466818668cd0019a09a0419ed4b9a002978a004cd19a0043477a0009a28017349
9a005cd266800cd00d002e71c519a06191484d001ba8ce6800cd2f5a2e01494c033eb4bda81077a38340067b5140c5a4a042e697b500252f4a000514005
25001d68c516002293148031450 0a2800e94b4005140841 4b40c43450021eb474ef400678a4a0028a005c73498a402d14005028014633d29d4f700c0f4
a6145f4a1a0431a3c534a62a2c55c4db8e69369a402e28e6818518a0031450210f141a004c525300c518a0031da8c628013148450210 9c500d031e0d34e
3d2810840a6b0a06376e2900cd300ce28c0c530023d29b8a00434629807a53a900bde940c520003de976d001b6976d0018e714d34000ffd9
0 client pkts, 5 server pkts, 0 turns.
137.30.120.37:80 → 137.30.123.234:2026 (156 kB)
Show as Raw
No delta times
Stream 1
Find: ffd9
Case sensitive
Find Next
Help
Filter Out This Stream
Print
Save as...
Back
Close
Activate Windows
Go to Settings to activate Windows.


Wireshark · Follow HTTP Stream (tcp.stream eq 1) · rhino2.log
1091ed4dea6800c521a003ea28a0038a38a004e28e0d0018a36d0018a3140c304d18c502131462800028a004c0f4a31cd0014714000c521a004cfb519a0
631c13d293073c9a0008cf06919698094d6cee142011c5201cd310bb4d0474e2800ed49fc340001d29c060f340c70e1b1430e38ea2900c5e7340fbdc531
1267de8a4327c525320334b4005140077a2800a3f1a0028a0033450014b9a0001a05001f5a3340094bd2801296801296800cd140c28a005079e9450014a
3ad0014ea0028a003a51de800a5a005c5148028a6003de8c50027d2945001d471498e2801693140062814 00bde8c5201314b8a60380e2908a401b68c530
0c51400a3d2948a003b51d6810b8a4c503171494085e281400b494c05a4a401466801791494000345001450 01f8d2628017145218 98ef41145803146280
0c518a0031462800db4628b0098f6e28c50018a4db458618a4c5200c52ed34006280050018e29769a761098f6a5e334862f14629885c514c033466810a2
8e2800a5cd300cd19a2e019e69290051400714 98e290c31450018a4c5001b4fad2104500216c75a5dc290ec2139a691cd00211cd18a0008a31c500260d2
127d281899a4a0028fc680147d69681098f7a5c63bd00263de8fc6800fc68e2800e3d69a6800e28a004eb476a062518a004a41fad0029c629314007149c
503138a300d000def4cdc3d29a108483499f7a620fc68c616801c071cd29148633386a796e29d8437071c534654f3400edd93d29d48658c514c80a2800a
280128a0033467de801334bcd0019a514005140064668a005ed41e940094500145001 4b40094a38a005a2800a05030e296801451de810629681852f1400
51f8d002d14000a2800a281052f140c4a3140051400514805a3bd001f852d0000e28ea680168a6018a3f1a0000a31400b8a39a0418a280173476a004ef4
b400514005140051400514051d2800a00a000d25002d140051400518a0028a2c02e29314c028fad200a2980525218628c5020a3038a2c30c5183400628da28b
0015e28c52b0060d18f6a2c0211da8a0029738ed4c05cd19a041c5045002629714006 28ef400b4845002d25002d140c4a2800cd19a402668a004c0a300d
031a569369070290060d001f4a0046e0f4a4cf3cd21867de90f3de8010ad1b680131460d001cd19f5a0039c525001cd1f85001cd140084d2648ed400527
b50018f4a4c66980bb7be68c1cd2002a6939ef4c04e73487340076a4c62800a4a0009a4e3d298084034dda298836d285c50161ffce8238a431bb0d382d1
710b8a4c668186df6a5c50058c52115440628eb4804a280128a003ad274a00051de81852839a0426ecd3a8001c8a5ef400b8a3b500211de931400514005
1400bd7f0a3a5002d34f5a005a5c60d030a514085e281d68017349ef40c5a2800a334085a3340c28a042e73450026734b40c28cd00267934b9e690051d6
80168cd0019e28a6028a322900a2834c0323346ee68106ee4500d00283466818668cd0019a09a0419ed4b9a002978a004cd19a0043477a0009a28017349
9a005cd266800cd00d002e71c519a06191484d001ba8ce6800cd2f5a2e01494c033eb4bda81077a38340067b5140c5a4a042e697b500252f4a000514005
25001d68c516002293148031450 0a2800e94b4005140841 4b40c43450021eb474ef400678a4a0028a005c73498a402d14005028014633d29d4f700c0f4
a6145f4a1a0431a3c534a62a2c55c4db8e69369a402e28e6818518a0031450210f141a004c525300c518a0031da8c628013148450210 9c500d031e0d34e
3d2810840a6b0a06376e2900cd300ce28c0c530023d29b8a00434629807a53a900bde940c520003de976d001b6976d0018e714d34000ffd9
0 client pkts, 5 server pkts, 0 turns.
137.30.120.37:80 → 137.30.123.234:2026 (156 kB)
Show as Raw
No delta times
Stream 1
Find: ffd8
Case sensitive
Find Next
Activate Windows


(root㉿kali)-[/home/kali/SBT-DF204-Case4]
okteta
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime
-root'
QStandardPaths: XDG_RUNTIME_DIR not set, defaulting to '/tmp/runtime
-root'

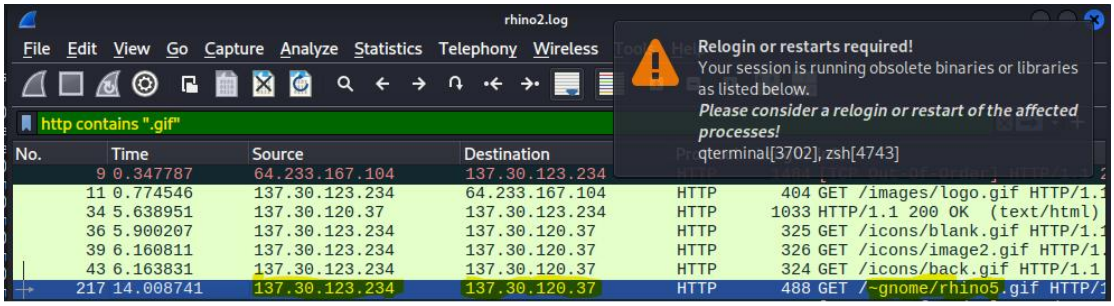

Save As
Name: rhino4.jpg
Relogin or restarts required!
Your session is running obsolete binaries or libraries as listed below.
Please consider a relogin or restart of the affected processes!
qterminal[3702], zsh[4743]
Home
Desktop
SBT-DF204-Case4
Other Locations
home
kali
SBT-DF204-Case4
Name
evidence
exported
recup_dir.1
reports
screenshots
steg
working
contraband.zip 230.6 kB Archive Wed
DFRWS2005-RODEO.zip 3.6 MB Archive 22 Oct
DFRWS2005-RODEO.zip.1 3.6 MB Text 22 Oct
DFRWS2005-RODEO.zip.2 809.1 kB Text Fri
DFRWS2005-RODEO.zip.3 3.6 MB Text Fri
gumbo1_output.txt 2.8 kB Text Tue
All files
Activate Windows
Cancel
Save

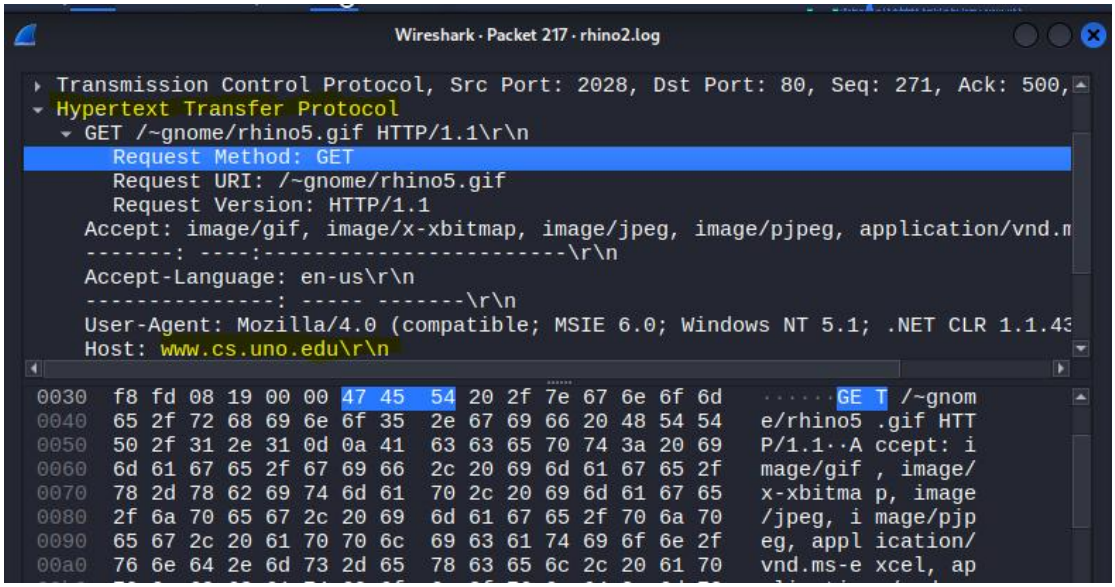

rhino4.jpg — Okteta
File Edit View Windows Bookmarks Tools Settings Help
New Open Save Save As Undo Redo Cut Copy Past
Fi&lesystem
Home
Name Size Date
rhino4.jpg
V&alue/Char Table
Dec Hex Oct Bin Char
000 00 000 00000000
001 01 001 00000001
002 02 002 00000010 .
003 03 003 00000011
004 04 004 00000100
005 05 005 00000101
006 06 006 00000110
007 07 007 00000111
008 08 010 00001000
009 09 011 00001001
Number (bytes): 1
F... Value/Ch... S... Ch... Str... Decodin...
Bi&nary Filter
Operation: operand AND data
Parameters
Operand: Hex
Align at end:

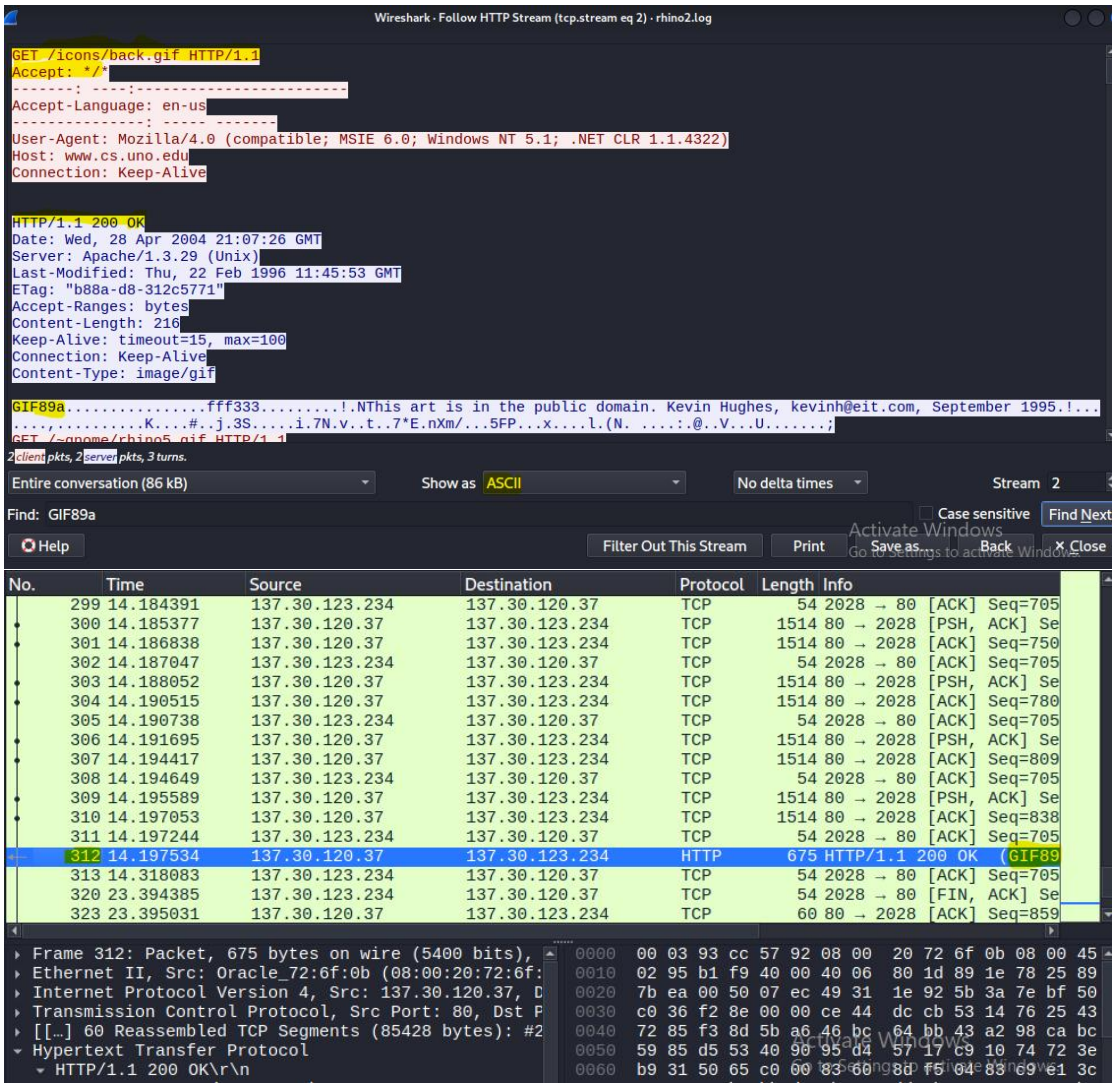
```
└─# xxd rhino4.jpg | head
00000000: 6666 6438 6666 6530 3030 3130 3461 3436  ffd8ffe000104a46
00000010: 3439 3436 3030 3031 3030 3031 3031 3263  4946000100010012c
00000020: 3031 3263 3030 3030 6666 6665 3030 3137  012c0000fffe0017
00000030: 3535 3264 3463 3635 3631 3634 3230 3533  552d4c6561642053
00000040: 3739 3733 3734 3635 3664 3733 3263 3230  797374656d732c20
00000050: 3439 3665 3633 3265 3030 6666 6462 3030  496e632e00ffdb00
00000060: 3834 3030 3432 3264 3332 3361 3332 3239  8400422d323a3229
00000070: 3432 3361 3336 3361 3462 3437 3432 3466  423a363a4b47424f
00000080: 3634 6137 3663 3634 3562 3562 3634 6363  64a76c645b5b64cc
00000090: 3932 3961 3739 6137 6632 6435 6665 6661  929a79a7f2d5fefa
```

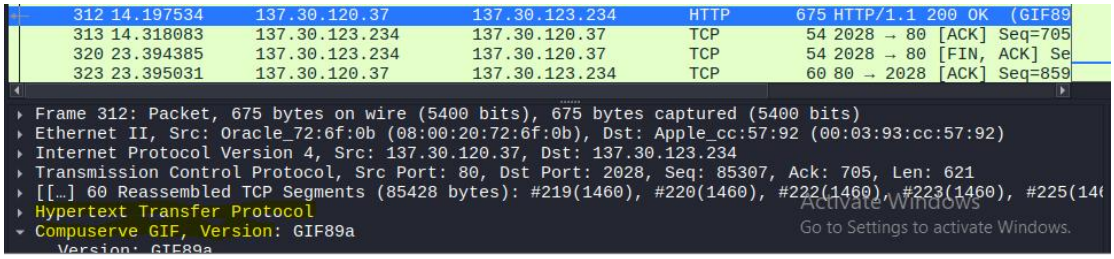
```
└─# xxd rhino4.jpg | tail
0004ac30: 3330 3063 3531 3861 3030 3331 6461 3863  300c518a0031da8c
0004ac40: 3632 3830 3133 3134 3834 3530 3231 3039  6280131484502109
0004ac50: 6335 3030 6430 3331 6530 6433 3465 3364  c500d031e0d34e3d
0004ac60: 3238 3130 3834 3061 3662 3061 3036 3337  2810840a6b0a0637
0004ac70: 3665 3239 3030 6364 3330 3063 6532 3863  6e2900cd300ce28c
0004ac80: 3063 3533 3030 3233 6432 3962 3861 3030  0c530023d29b8a00
0004ac90: 3433 3436 3239 3830 3761 3533 6139 3030  434629807a53a900
0004aca0: 6264 6539 3430 6335 3230 3030 3364 6539  bde940c520003de9
0004acb0: 3736 6430 3031 6236 3937 3664 3030 3138  76d001b6976d0018
0004acc0: 6537 3134 6433 3430 3030 6666 6439 0a    e714d34000ffd9.
```

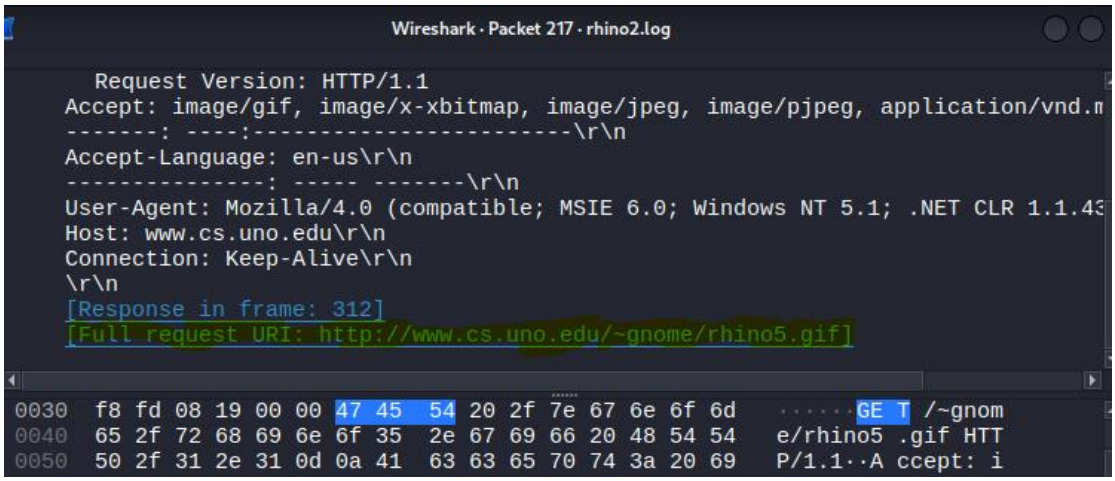
```
┌──(root㉿kali)-[/home/kali/SBT-DF204-Case4]
└─# display rhino4.jpg
```

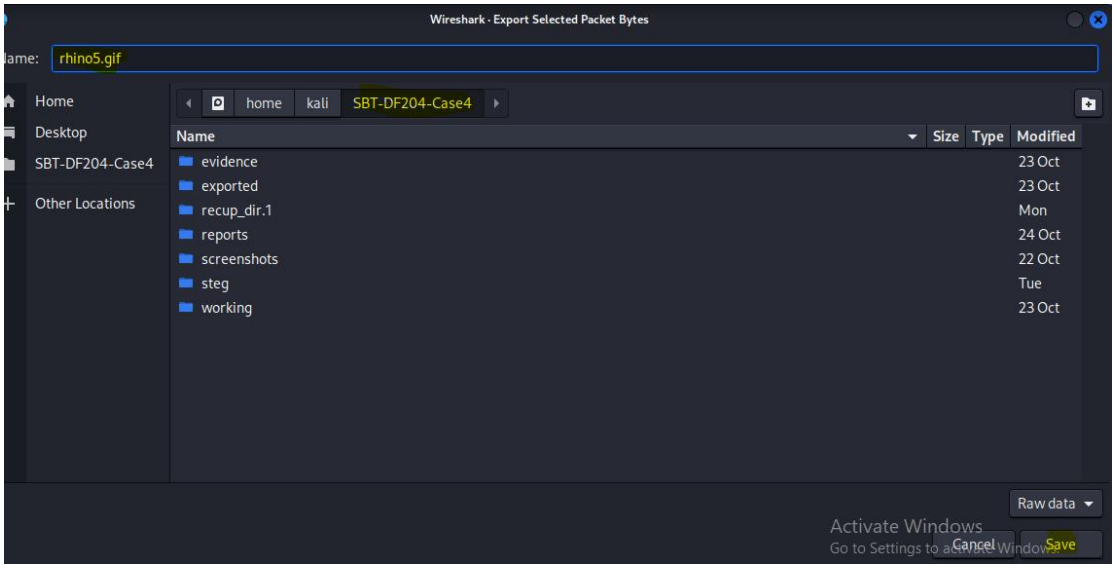


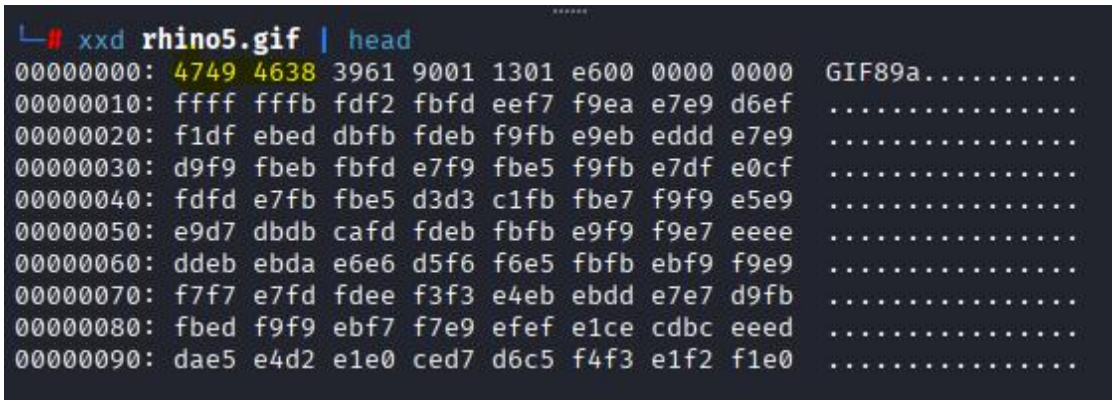

Wireshark · Packet 217 · rhino2.log
Transmission Control Protocol, Src Port: 2028, Dst Port: 80, Seq: 271, Ack: 500,
Hypertext Transfer Protocol
GET /~gnome/rhino5.gif HTTP/1.1\r\n
Request Method: GET
Request URI: /~gnome/rhino5.gif
Request Version: HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/vnd.m
-------: ----:-------------------------\r\n
Accept-Language: en-us\r\n
---------------: ----- -------\r\n
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.43
Host: www.cs.uno.edu\r\n
0030 f8 fd 08 19 00 00 47 45 54 20 2f 7e 67 6e 6f 6d ····GE T /~gnom
0040 65 2f 72 68 69 6e 6f 35 2e 67 69 66 20 48 54 54 e/rhino5 .gif HTT
0050 50 2f 31 2e 31 0d 0a 41 63 63 65 70 74 3a 20 69 P/1.1··A ccept: i
0060 6d 61 67 65 2f 67 69 66 2c 20 69 6d 61 67 65 2f mage/gif , image/
0070 78 2d 78 62 69 74 6d 61 70 2c 20 69 6d 61 67 65 x-xbitma p, image
0080 2f 6a 70 65 67 2c 20 69 6d 61 67 65 2f 70 6a 70 /jpeg, i mage/pjp
0090 65 67 2c 20 61 70 70 6c 69 63 61 74 69 6f 6e 2f eg, appl ication/
00a0 76 6e 64 2e 6d 73 2d 65 78 63 65 6c 2c 20 61 70 vnd.ms-e xcel, ap

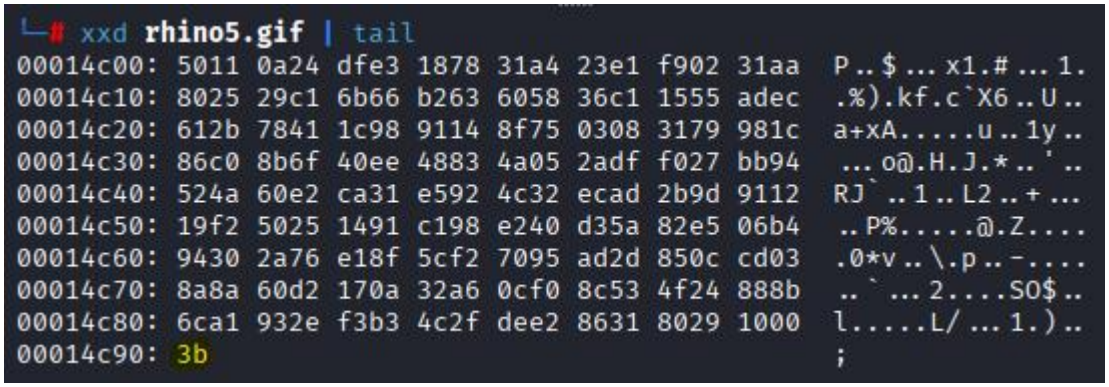

Wireshark · Follow HTTP Stream (tcp.stream eq 2) · rhino2.log
GET /icons/back.gif HTTP/1.1
Accept: */*
-------: ----:----------------------
Accept-Language: en-us
---------------: ----- -------
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.4322)
Host: www.cs.uno.edu
Connection: Keep-Alive
HTTP/1.1 200 OK
Date: Wed, 28 Apr 2004 21:07:26 GMT
Server: Apache/1.3.29 (Unix)
Last-Modified: Thu, 22 Feb 1996 11:45:53 GMT
ETag: "b88a-d8-312c5771"
Accept-Ranges: bytes
Content-Length: 216
Keep-Alive: timeout=15, max=100
Connection: Keep-Alive
Content-Type: image/gif
GIF89a.....fff333.....!.NThis art is in the public domain. Kevin Hughes, kevinh@eit.com, September 1995.!...
...,.....K....#..j.3S.....i.7N.v..t..7*E.nXm/...5FP...x....l.(N. ....:.@..V...U.....;
GET /~gnome/rhino5.gif HTTP/1.1
2 client pkts, 2 server pkts, 3 turns.
Entire conversation (86 kB)
Show as ASCII
No delta times
Stream 2
Find: GIF89a
Case sensitive
Find Next
Help
Filter Out This Stream
Print
Save as...
Back
Close
No. Time Source Destination Protocol Length Info
299 14.184391 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [ACK] Seq=705
300 14.185377 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [PSH, ACK] Se
301 14.186838 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [ACK] Seq=750
302 14.187047 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [ACK] Seq=705
303 14.188052 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [PSH, ACK] Se
304 14.190515 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [ACK] Seq=780
305 14.190738 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [ACK] Seq=705
306 14.191695 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [PSH, ACK] Se
307 14.194417 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [ACK] Seq=809
308 14.194649 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [ACK] Seq=705
309 14.195589 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [PSH, ACK] Se
310 14.197053 137.30.120.37 137.30.123.234 TCP 1514 80 → 2028 [ACK] Seq=838
311 14.197244 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [ACK] Seq=705
312 14.197534 137.30.120.37 137.30.123.234 HTTP 675 HTTP/1.1 200 OK (GIF89
313 14.318083 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [ACK] Seq=705
320 23.394385 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [FIN, ACK] Se
323 23.395031 137.30.120.37 137.30.123.234 TCP 60 80 → 2028 [ACK] Seq=859
Frame 312: Packet, 675 bytes on wire (5400 bits),
Ethernet II, Src: Oracle_72:6f:0b (08:00:20:72:6f:
Internet Protocol Version 4, Src: 137.30.120.37, D
Transmission Control Protocol, Src Port: 80, Dst P
[[…] 60 Reassembled TCP Segments (85428 bytes): #2
Hypertext Transfer Protocol
HTTP/1.1 200 OK\r\n
0000 00 03 93 cc 57 92 08 00 20 72 6f 0b 08 00 45
0010 02 95 b1 f9 40 00 40 06 80 1d 89 1e 78 25 89
0020 7b ea 00 50 07 ec 49 31 1e 92 5b 3a 7e bf 50
0030 c0 36 f2 8e 00 00 ce 44 dc cb 53 14 76 25 43
0040 72 85 f3 8d 5b a6 46 bc 64 bb 43 a2 98 ca bc
0050 59 85 d5 53 40 90 95 d4 57 17 c9 10 74 72 3e
0060 b9 31 50 65 c0 00 83 60 8d f5 04 83 69 e1 3c

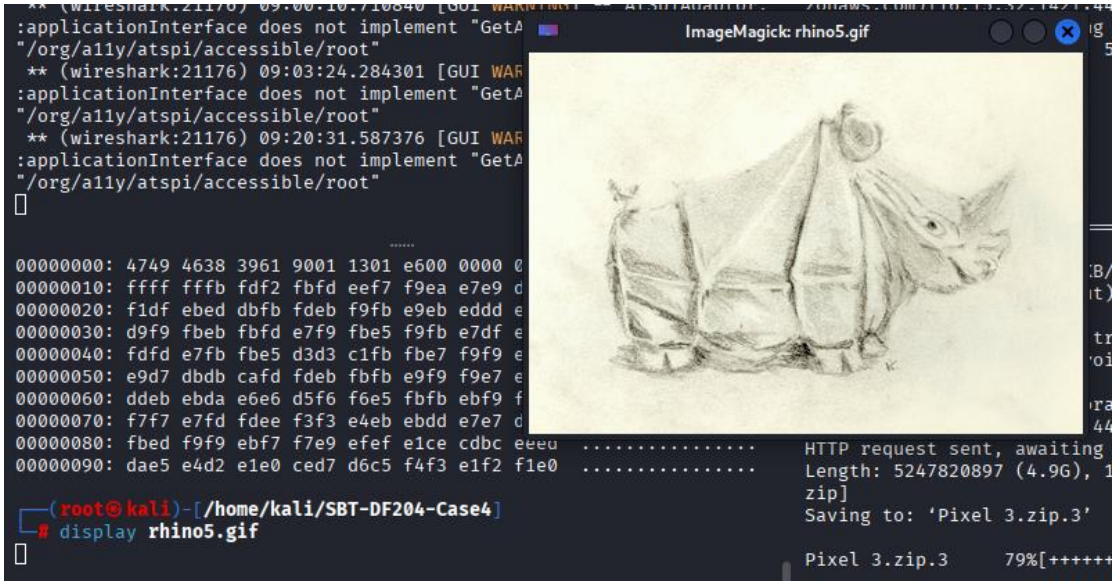

312 14.197534 137.30.120.37 137.30.123.234 HTTP 675 HTTP/1.1 200 OK (GIF89
313 14.318083 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [ACK] Seq=705
320 23.394385 137.30.123.234 137.30.120.37 TCP 54 2028 → 80 [FIN, ACK] Se
323 23.395031 137.30.120.37 137.30.123.234 TCP 60 80 → 2028 [ACK] Seq=859
Frame 312: Packet, 675 bytes on wire (5400 bits), 675 bytes captured (5400 bits)
Ethernet II, Src: Oracle_72:6f:0b (08:00:20:72:6f:0b), Dst: Apple_cc:57:92 (00:03:93:cc:57:92)
Internet Protocol Version 4, Src: 137.30.120.37, Dst: 137.30.123.234
Transmission Control Protocol, Src Port: 80, Dst Port: 2028, Seq: 85307, Ack: 705, Len: 621
[[…] 60 Reassembled TCP Segments (85428 bytes): #219(1460), #220(1460), #222(1460), #223(1460), #225(14
Hypertext Transfer Protocol
Compuserve GIF, Version: GIF89a
Version: GIF89a
Activate Windows
Go to Settings to activate Windows.


Wireshark · Packet 217 · rhino2.log
Request Version: HTTP/1.1
Accept: image/gif, image/x-xbitmap, image/jpeg, image/pjpeg, application/vnd.m
-------: ----:------------------------\r\n
Accept-Language: en-us\r\n
---------------: ----- -------\r\n
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; .NET CLR 1.1.43
Host: www.cs.uno.edu\r\n
Connection: Keep-Alive\r\n
\r\n
[Response in frame: 312]
[Full request URI: http://www.cs.uno.edu/~gnome/rhino5.gif]
0030 f8 fd 08 19 00 00 47 45 54 20 2f 7e 67 6e 6f 6d ·····GE T /~gnom
0040 65 2f 72 68 69 6e 6f 35 2e 67 69 66 20 48 54 54 e/rhino5 .gif HTT
0050 50 2f 31 2e 31 0d 0a 41 63 63 65 70 74 3a 20 69 P/1.1··A ccept: i


Wireshark · Export Selected Packet Bytes
Name: rhino5.gif
Home
Desktop
SBT-DF204-Case4
Other Locations
home kali SBT-DF204-Case4
Name Size Type Modified
evidence 23 Oct
exported 23 Oct
recup_dir.1 Mon
reports 24 Oct
screenshots 22 Oct
steg Tue
working 23 Oct
Raw data
Cancel
Save
Activate Windows
Go to Settings to activate Windows.


└─# xxd rhino5.gif | head
00000000: 4749 4638 3961 9001 1301 e600 0000 0000 GIF89a.....
00000010: ffff fffb fdf2 fbfd eef7 f9ea e7e9 d6ef ....
00000020: f1df ebed dbfb fdeb f9fb e9eb eddd e7e9 ....
00000030: d9f9 fbeb fbfd e7f9 fbe5 f9fb e7df e0cf ....
00000040: fdfd e7fb fbe5 d3d3 c1fb fbe7 f9f9 e5e9 ....
00000050: e9d7 dbdb cafd fdeb fbfb e9f9 f9e7 eeee ....
00000060: ddeb ebda e6e6 d5f6 f6e5 fbfb ebf9 f9e9 ....
00000070: f7f7 e7fd fdee f3f3 e4eb ebdd e7e7 d9fb ....
00000080: fbed f9f9 ebf7 f7e9 efef e1ce cdbc eeed ....
00000090: dae5 e4d2 e1e0 ced7 d6c5 f4f3 e1f2 f1e0 ....





Item	Findings
Material request(s): method, URI, host	GET /gnome/rhino4.jpg HTTP/1.1 (packet 49) and GET /~gnome/rhino5.gif HTTP/1.1 (packet 217). Host confirmed in packet 217's headers: www.cs.uno.edu, both requests target the same gnome user directory seen earlier in the FTP session
Client and server endpoints	Client: 137.30.123.234 (port 2028 in packet 217). Server: 137.30.120.37 (port 80, standard HTTP)
TCP stream / packet range	Packets 34 (unrelated 200 OK, likely a preceding page load), 39, 43 (icon/image requests, not case-relevant), **49** (rhino4.jpg request) and **217** (rhino5.gif request), both

Item	Findings
	case-relevant requests within the same capture, 487 packets/~14 seconds apart
Response code and content type	Packet 34 confirms the server responding 200 OK with text/html content type for an earlier page request in the same session; the specific response packets for the rhino4.jpg and rhino5.gif GET requests themselves aren't captured in the filtered view shown, so the actual transfer status for those two files was confirmed by following the full TCP stream for packets 49 and 217 (**Follow → HTTP Stream**)
Exported object: type, size, hash	rhino5.gif. MD5 1e90b7f70b2ecb605898524a88269029 (both the HTTP-exported copy and the on-disk working copy)
Matches another artefact in the case? (compare hashes)	**Confirmed match, identical hash.** exported_http_rhino2/rhino5.gif and the rhino5.gif already in the case working directory share the exact same MD5 (1e90b7f70b2ecb605898524a88269029). This **conclusively proves** the file on disk is the same file transferred over HTTP in this capture, byte for byte.
Client identification	User-Agent string identifies the requesting client as **Internet Explorer 6.0, running on Windows XP** (Windows NT 5.1)
Authentic rhino4.jpg (HTTP-exported)	Valid JPEG, JFIF 1.00, **1531×2286**, 153 kB, 300×300 DPI, **Comment: "U-Lead Systems, Inc."**, baseline encoding, 3 color components. Hex-verified: begins FFD8 FFE0, a genuine, structurally valid JPEG header

9. Eight Investigation Questions - Answers

1. How did you preserve the case materials?
All original evidence files (`RHINOUSB.dd`, `rhino.log`, `rhino2.log`, `rhino3.log`) were hashed with both MD5 and SHA-256 immediately upon receipt to establish a baseline. A working copy of the disk image was created using `dd` (bit-for-bit duplication), and its hash was verified identical to the original (`sha256sum` match confirmed) before any analysis began. The original files were kept untouched; all carving, extraction, and password-cracking work was performed exclusively against the working copy.

2. What does the USB disk image show before recovery?
`RHINOUSB.dd` is a 259,506,176-byte (260 MB) raw image containing a single **FAT16** file system (`mkdosfs`-formatted, Volume ID `0x4092d9d1`), unpartitioned at the MBR level (confirmed via `fdisk/parted`). `fls` listing showed only two allocated files at the time of imaging `gumbo1.txt` and `gumbo2.txt`, with no deleted entries visible in the active directory structure.

3. What deleted or recoverable files are supported?
PhotoRec carving of the full disk image recovered **132 files** from unallocated space, including numerous JPEGs, GIFs, text fragments, and a `.doc` file, none of which were visible in the live FAT16 directory structure. This confirms substantial deleted/unallocated content existed on the device beyond what the active file system exposed, including at least one image (`f0106409.jpg`) that matches evidence recovered independently elsewhere in the case.

4. Is there evidence of hidden content in the recovered media?
Yes , confirmed via `stegdetect/stegbreak`. Two recovered JPEGs tested positive for **JPHide v5** steganography: `f0105065.jpg` (passphrase cracked: `gator`) and `f0104249.jpg` (passphrase cracked: `gumbo`), both cracked via dictionary attack against `rockyou.txt`. Hidden image content was successfully extracted from both using `jpseek.exe`, producing two valid, viewable JPEGs (`r065.jpg`, `r249.jpg`) with embedded EXIF data showing original capture dates predating the carrier files' modification dates, consistent with deliberate concealment.

5. What does the FTP capture establish?
`rhino.log` captures a user authenticating via FTP as `gnome` (password `gnome123`) from server `137.30.122.253`, who then **uploaded** `rhino1.jpg` (65,703 bytes) to the remote host in binary mode. This matches the `rhino1.jpg` already present in the case evidence, establishing that this file was actively transmitted off the source system during the captured session.

6. What does the protected archive analysis establish?
`contraband.zip` was a password-protected archive concealing a single file, `rhino2.jpg`. The weak password `monkey` was cracked via `fcrackzip` using the `rockyou.txt` wordlist. The extracted `rhino2.jpg` (230,665 bytes) matches, by file size, a JPEG independently carved from unallocated disk space (`f0106409.jpg`), indicating the same image existed on the device both as deliberately-hidden archived content and as residual deleted data.

7. What does the HTTP capture establish?
`rhino2.log` captures the same client (`137.30.123.234`, confirmed via WHOIS as belonging to the University of New Orleans) retrieving two files via HTTP GET from `www.cs.uno.edu/~gnome/`: `rhino4.jpg` and `rhino5.gif`, using Internet Explorer 6.0 on Windows XP. Objects exported directly from this capture were hash-compared against on-disk case files: `rhino5.gif` matched exactly (MD5 `1e90b7f70b2ecb605898524a88269029`), confirming authenticity, while `rhino4.jpg` **did not match,** the on-disk copy is invalid/corrupted, and the HTTP-exported copy (a genuine 1531×2286 JPEG, authored with U-Lead Systems software) is the authentic artefact.

8. What conclusion can you defend?
The evidence establishes a consistent pattern of activity by a single user (`gnome`), operating across network protocols (FTP upload, HTTP retrieval) and local concealment methods (password-protected archives, JPHide steganography) at IP addresses attributable to the University of New Orleans. Multiple independently-recovered artefacts corroborate one another, files carved from unallocated space match files extracted from encrypted archives and files captured in transit over the network, establishing a high-confidence, cross-validated timeline of file handling and concealment. Separately, a **chain-of-custody concern** was identified and should be formally noted: the on-disk `rhino4.jpg` in the case working directory does not match the authentic file established via network capture, indicating file substitution, corruption, or mishandling occurred at some point after acquisition and must be accounted for before this evidence set is considered complete and defensible.

10. Executive Summary

This investigation examined a USB storage device image (RHINOUSB.dd), three network packet captures (rhino.log, rhino2.log, rhino3.log), and a password-protected archive (contraband.zip), using standard digital forensics tools (The Sleuth Kit, PhotoRec, Wireshark, stegdetect/stegbreak, fcrackzip) under a documented chain of custody with hash-verified evidence integrity throughout.

Key findings:

- The USB device's FAT16 file system exposed only two allocated files, but carving unallocated space with PhotoRec recovered **132 additional files**, revealing substantial deleted content not visible through normal file browsing.
- **Steganography was confirmed and defeated**: two recovered JPEGs were found to contain JPHide-concealed data. Both were cracked via dictionary attack (passphrases gator and gumbo) and the hidden images successfully extracted and verified as valid, viewable photographs with EXIF metadata showing they were captured years before the carrier files were last modified, consistent with deliberate concealment.
- **A password-protected archive was defeated**: contraband.zip's weak password (monkey) was cracked in seconds, revealing rhino2.jpg, an image that independently matches, by size, a file separately recovered from unallocated disk space, corroborating that this content existed on the device through two distinct, independently-verifiable paths.
- **Network activity was reconstructed and attributed**: both an FTP upload and HTTP downloads were traced to the same user account (gnome) and the same source IP range, which WHOIS attributes to the University of New Orleans. One HTTP-retrieved file (rhino5.gif) was hash-confirmed identical to its on-disk counterpart, validating the integrity of that evidence chain.
- **A significant integrity issue was identified**: the on-disk copy of rhino4.jpg in the case working directory does **not** match the authentic file established via the HTTP packet capture, the stored file is corrupted or has been substituted. This was independently flagged earlier by a failed JPEG signature check and confirmed conclusively by hash mismatch once the genuine file was recovered from the network capture.

The combined evidence carved files, cracked archives, defeated steganography, and reconstructed network sessions, paints a consistent, cross-corroborated picture of a single user's activity across multiple protocols and concealment techniques, all traceable to the same network origin. That said, the rhino4.jpg discrepancy is a legitimate evidence-handling concern that must be disclosed and accounted for rather than quietly resolved, since it affects the defensibility of any conclusion drawn from that specific file. With that caveat flagged, the remaining findings are well-supported and independently verifiable.